StellarisWare Release Notes



SW-RLN-6288

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Revision Information

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1.1 Bug Fixes in Stellaris Boot Loader

1.1.1 CAN boot loader did not work on Fury-class devices (Reference 12270)

The CAN boot loader did not account for the differences in the CAN interfaces on Fury-class devices (the need to run from the PLL, the fixed 8 MHz input clock to the CAN module, the paced writes, and the delayed reads). There is a new configuration option, CAN_REQUIRES_PLL, that makes the appropriate adjustments to allow the CAN boot loader to operation properly on Fury-class devices.

1.2 Bug Fixes in Stellaris Peripheral Driver Library

1.2.1 CANBitRateSet produces invalid bit timings (Reference 12455)

The CANBitRateSet API would produce invalid CAN bit timings for certain input clock and CAN baud rate combinations. This has been corrected.

1.3 New Features in Stellaris Graphics Library

1.3.1 Improve NumLeadingZeros macro in graphics library (for CCS) (Reference 12219)

The graphics library string module has a macro named NumLeadingZeros. The implementation of the macro was improved for the CCS toolchain. There is no change for other toolchains.

1.4 Bug Fixes in Stellaris USB Library

1.4.1 Memory allocation issue with USB FIFO configuration data (Reference 12266)

The default size of the memory allocated to the USB FIFO configuration memory could only hold information on three IN or three OUT endpoints. If more than three of either type of endpoint was used, the USB library would overflow this memory allocation and begin reading and writing to memory beyond the space allocated to the USB FIFO configuration. This under allocation causes issues with any USB device that uses more than three IN or three OUT endpoints and is most likely to occur when creating a custom composite device where the total number of IN or OUT endpoints is greater than three. To fix this, the default allocation was increased to 16 entries which is the maximum number of endpoints on any USB controller.

1.5 New Features in Stellaris Utility Library

1.5.1 Added Software SSI module (Reference 12269)

Added a software SSI module (SoftSSI) that allows an arbitrary set of GPIO pins to be used as a SPI master. Supports SPI modes 0-3, 4-16 data bits, application-configurable GPIO usage, and application-configurable FIFO size.

1.5.2 Added Software I2C module (Reference 12360)

Added a software I2C module (SoftI2C) that allows an arbitrary pair of GPIO pins to be used as an I2C master. Supports I2C master transmit and receive, and application-configurable GPIO usage.

1.5.3 Added cosine macro (Reference 12248)

Added a cosine macro to sine.h that simply calls the sine function with the angle incremented by 90 degrees.

1.6 New Features in DK-LM3S9B96 Firmware Package

1.6.1 Touchscreen driver handling of unrecognized daughter boards fixed. (Reference 12359)

In previous releases, the touchscreen driver did not correctly default to “normal” behavior when it encountered an unrecognized daughter board ID. The code has been reworked to ensure that an unknown daughter board ID is handled in the same way as the SDRAM daughter board or cases where no daughter board is present. The assumption is that any unknown daughter board will not have rewired the touchscreen interface (since this would require other changes in the touchscreen driver).

1.7 Bug Fixes in DK-LM3S9B96 Firmware Package

1.7.1 Echo mode in usb_bulk_example corrected (Reference 12335)

A recent buffer size change in the Windows “usb_bulk_example” application had the inadvertent side effect of preventing echo mode (“-e” command line switch) from working correctly. This has now been corrected and echo mode operates as expected.

1.7.2 Fixed dbeeprom example to terminate strings correctly (Reference 12347)

The dbeeprom example application that is used to read and write the content of the ID EEPROM on daughterboards failed to terminate the board description string correctly. This has now been fixed. Also added support for writing the required ID information for the new EM2 daughterboard.

1.7.3 USB DFU driver version numbering corrected (Reference 12427)

The version numbering of the USB DFU driver DLL, lmdfu.dll was incorrect in the previous release and this caused problems for some people attempting to update the driver. This has now been fixed and the DLL version number correctly tracks the release number.

1.7.4 i2s_filter example can fail and generate loud noise (Reference 11694)

The i2s_filter example can generate loud noise rather than streaming the audio from the line input to the audio outputs when using a debugger. The sound driver (drivers/sound.c) is separately enabling the I2S transmit and receive interfaces which can cause the interfaces to become out of sync when using a debugger. If a receive channel is being used, the new sound driver will enable both transmit and receive at the same by calling the I2STxRxEnable() function instead of the individual I2STxEnable() and I2SRxEnable() functions.

1.8 Bug Fixes in EK-LM3S2965 Firmware Package

1.8.1 can_fifo example was not clearing the FIFO flag properly (Reference 12316)

The can_fifo example was not clearing the MSG_OBJ_FIFO bit in the last message object that was used in the list of CAN message objects that made up the FIFO. The can_fifo example now insures that the MSG_OBJ_FIFO bit is clear for the last message object in the CAN FIFO.

1.9 Bug Fixes in EK-LM3S3748 Firmware Package

1.9.1 Echo mode in usb_bulk_example corrected (Reference 12335)

A recent buffer size change in the Windows “usb_bulk_example” application had the inadvertent side effect of preventing echo mode (“-e” command line switch) from working correctly. This has now been corrected and echo mode operates as expected.

1.9.2 USB DFU driver version numbering corrected (Reference 12427)

The version numbering of the USB DFU driver DLL, lmdfu.dll was incorrect in the previous release and this caused problems for some people attempting to update the driver. This has now been fixed and the DLL version number correctly tracks the release number.

1.10 Bug Fixes in EK-LM3S8962 Firmware Package

1.10.1 can_fifo example was not clearing the FIFO flag properly (Reference 12316)

The can_fifo example was not clearing the MSG_OBJ_FIFO bit in the last message object that was used in the list of CAN message objects that made up the FIFO. The can_fifo example now insures that the MSG_OBJ_FIFO bit is clear for the last message object in the CAN FIFO.

1.11 Bug Fixes in EK-LM3S9B90 Firmware Package

1.11.1 Echo mode in usb_bulk_example corrected (Reference 12335)

A recent buffer size change in the Windows “usb_bulk_example” application had the inadvertent side effect of preventing echo mode (“-e” command line switch) from working correctly. This has now been corrected and echo mode operates as expected.

1.11.2 USB DFU driver version numbering corrected (Reference 12427)

The version numbering of the USB DFU driver DLL, lmdfu.dll was incorrect in the previous release and this caused problems for some people attempting to update the driver. This has now been fixed and the DLL version number correctly tracks the release number.

1.12 Bug Fixes in EK-LM3S9B92 Firmware Package

1.12.1 Echo mode in usb_bulk_example corrected (Reference 12335)

A recent buffer size change in the Windows “usb_bulk_example” application had the inadvertent side effect of preventing echo mode (“-e” command line switch) from working correctly. This has now been corrected and echo mode operates as expected.

1.12.2 USB DFU driver version numbering corrected (Reference 12427)

The version numbering of the USB DFU driver DLL, lmdfu.dll was incorrect in the previous release and this caused problems for some people attempting to update the driver. This has now been fixed and the DLL version number correctly tracks the release number.

2 Release Notes for StellarisWare Revision 6075 (June 4, 2010)

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2.1 New Features in Stellaris Boot Loader

2.1.1 Added support to cooperate with the in-Flash errata workaround (Reference 12088)

Certain revisions of some Stellaris microcontrollers come with a non-erasable errata workaround pre-programmed into Flash. Support has been added to the boot loader to allow it to be configured to work in these parts when located at 0x1000, the first portion of Flash that is available for customer use. This support is enabled via the WORKAROUND_COOP define in bl_config.h.

2.2 New Features in Stellaris Peripheral Driver Library

2.2.1 Function CPUprimask() added to cpu.c (Reference 12214)

A new function, CPUprimask(), has been added to the low level API offered by cpu.c. This function returns the current contents of the PRIMASK register and can be used to determine whether inter-

rupts are enabled or disabled at the CPU level. A non-zero return code indicates that interrupts are currently disabled.

2.3 Bug Fixes in Stellaris Peripheral Driver Library

2.3.1 USB_EP_HOST_IN and USB_EP_DEV_OUT definitions overlap with USB_EP_SPEED_FULL definition (Reference 12152)

The USB_EP_HOST_IN and USB_EP_DEV_OUT values had the same bit field position in the bit flags as the USB_EP_SPEED_FULL flag causing an overlap that resulted in errors when these flags were used. The functions affected were the USBFIFOFlush(), USBEndpointDMAEnable(), and USBEndpointDMADisable(). The USB_EP_HOST_IN and USB_EP_DEV_OUT are now simply the bitwise NOT of the USB_EP_HOST_OUT and USB_EP_DEV_IN to remove this collision and all uses of these bits have been changed to match this definition.

2.3.2 USBDevEndpointStallClear() is not clearing data toggle for non-zero endpoints (Reference 12165)

When the USBDevEndpointStallClear() function is called for non-zero endpoints and the ulFlags parameter is USB_EP_DEV_OUT, the function sets the wrong bit and does not clear the data toggle. This error causes the USB controller to ignore valid DATA0 packets after the USBDevEndpointStallClear() is called. This has been fixed and now the USBDevEndpointStallClear() function properly clears the data toggle when called.

2.3.3 Deprecated INT_ADC? from hw_ints.h (Reference 12203)

The defines for INT_ADC? (the four sample sequence interrupts for ADC0) have been deprecated in favor of INT_ADC0SS?. This makes them consistent with the sample sequence interrupt defines for ADC1, which are INT_ADC1SS?.

2.4 New Features in Stellaris Graphics Library

2.4.1 Added monospace font support to ftrasterize (Reference 10690)

Support has been added to ftrasterize to render a font with uniform spacing (in other words, a monospaced font). Additionally, renderings of the Computer Modern monospaced font in sizes 12 through 48 have been added to the default font collection for the Stellaris Graphics Library.

2.5 Bug Fixes in Stellaris USB Library

2.5.1 USB Host MSC class not handling STALLed commands properly (Reference 12105)

When a device issued a STALL on a SCSI command, the USB library improperly returned without requesting status from the device. This affected mostly larger USB MSC devices that would STALL commands before they were ready to respond to commands. This caused the USB host controller and the device to become out of sync and fail to communicate properly after the STALL condition. The USB host MSC class now properly requests status after any STALL on a SCSI command.

2.5.2 USBDAudioTerm() and USBDMSCTerm() do not call USBDCDTerm() (Reference 11676)

USBDAudioTerm() and USBDMSCTerm() were not calling USBDCDTerm() and were leaving the device class active after returning. USBDAudioTerm() and USBDMSCTerm() now call USBDCDTerm() before returning to disable USB device mode.

2.5.3 USBHCDClearFeature() was not resetting data toggle (Reference 12166)

When the USB library issued a Clear Feature request to clear the halt condition on an endpoint, it failed to also clear the data toggle on the endpoint. This caused the host controller to become out of sync with the device. The USBHCDClearFeature() function now properly clears the data toggle when issuing a Clear Feature Endpoint Halt request.

2.6 New Features in Third Party Packages

2.6.1 Updated to lwIP 1.3.2 (Reference 12145)

lwIP 1.3.2 has been incorporated into StellarisWare and all example applications that use lwIP have been updated to use the new version.

2.7 New Features in DK-LM3S9B96 Firmware Package

2.7.1 New Speex Encode/Decode example added (Reference 11069)

A new example was added that performs real-time Speex encode and decode with an incoming audio signal on the line input.

2.7.2 USB Drivers Now Support Windows 7 32-bit and 64-bit Versions (Reference 11555)

The SW-USB-windrivers-xxxx.zip USB driver package has been updated to support both 32-bit and 64-bit versions of Windows 7. The package includes 64-bit versions of URLs lmusbdll.dll and lmdfu.dll allowing 64-bit applications to access these interfaces. The Device Firmware Upgrade driver has also been rewritten to use the Microsoft-supplied WINUSB interface rather than libusb-win32 which is not currently fully supported on Windows 7.

2.7.3 Add DMA support to the USB audio device examples (Reference 11677)

In previous releases, the USB device audio examples did not use uDMA for transfers. The stand-alone usb_dev_audio and usb_dev_caudiohid examples now both use uDMA for transfers.

2.7.4 Windows tool Visual Studio projects updated to VS2008 (Reference 12138)

The project files included in the tools directory for the various VisualStudio applications and DLLs have been updated to support VisualStudio2008. They were previously for VisualStudio2005.

2.7.5 New dbeeprom example application added (Reference 11936)

A new example application, dbeeprom, has been added to the dk-lm3s9b96 StellarisWare release. This allows the contents of the ID EEPROM on option daughterboards to be read and written and is intended to allow recovery from cases where a user application accidentally corrupts or erases the content of this device. The application provides a command-line interface via UART0 and is capable of writing the ID block for both the SRAM/Flash/LCD and FPGA/Camera/LCD daughterboards.

2.8 Bug Fixes in DK-LM3S9B96 Firmware Package

2.8.1 Allow SafeRTOS demo to be moved from location 0 (Reference 12086)

The vector table address in the xPORT_INIT_PARAMETERS structure that is passed to SafeRTOS was hard coded to 0, which prevented the SafeRTOS scheduler from starting if the vector table was not actually located at 0 (for example, when using a boot loader). That address is now filled in at run time from the NVIC vector table base address register, therefore reflecting the actual vector table in use. This does not affect the operation of the demo as it is provided, but does make it possible to change the base address of the demo (in the linker script) and have it continue to work correctly.

2.8.2 usb_host_audio application does not display time information (Reference 11974)

The usb_host_audio application did not properly compute the duration in minutes and seconds for wav files that were selected. The application also failed to update the elapsed time as a file was played. Both of these issues were fixed.

2.8.3 usb_stick_update would not enumerate drives in all cases. (Reference 12241)

The usb_stick_update only enumerated USB MSC devices if they were present when the application started. The usb_stick_update now properly enumerates devices by forcing the OTG controller to indicate a host connection which allows USB MSC devices to properly enumerate.

2.8.4 Fixed left/right channel swap on I2S DAC output (Reference 12245)

The left and right channels were swapped on the I2S DAC outputs being sent to both the headphone and line out jacks on the board. The DAC now swaps the two channels so that they appear correctly on the headphone and line out jacks.

2.9 Bug Fixes in EK-LM3S1968 Firmware Package

2.9.1 RIT display driver used wrong SPI mode (Reference 12217)

The display driver for the RIT 128x96 panel on the evaluation board used the wrong SPI mode. This caused the Stellaris SPI port to transition the data line on the same clock edge that the display controller read the data line. While this managed to work, the correct SPI mode is now used. Additionally, some improvements were made in the handling of the SPI interface, resulting in better performance of the display driver.

2.9.2 Added MPU region to mpu_fault example for bit-banded SRAM (Reference 12235)

A new MPU region has been added to the mpu_fault example to allow read/write access to bit-banded SRAM. This is required by the updated display driver, which uses the bit-banded SRAM region for storing flags.

2.10 Bug Fixes in EK-LM3S2965 Firmware Package

2.10.1 RIT display driver used wrong SPI mode (Reference 12217)

The display driver for the RIT 128x96 panel on the evaluation board used the wrong SPI mode. This caused the Stellaris SPI port to transition the data line on the same clock edge that the display controller read the data line. While this managed to work, the correct SPI mode is now used. Additionally, some improvements were made in the handling of the SPI interface, resulting in better performance of the display driver.

2.10.2 Added MPU region to mpu_fault example for bit-banded SRAM (Reference 12235)

A new MPU region has been added to the mpu_fault example to allow read/write access to bit-banded SRAM. This is required by the updated display driver, which uses the bit-banded SRAM region for storing flags.

2.11 New Features in EK-LM3S3748 Firmware Package

2.11.1 USB Drivers Now Support Windows 7 32-bit and 64-bit Versions (Reference 11555)

The SW-USB-windrivers-xxxx.zip USB driver package has been updated to support both 32-bit and 64-bit versions of Windows 7. The package includes 64-bit versions of URLs lmusbdll.dll and lmdfu.dll allowing 64-bit applications to access these interfaces. The Device Firmware Upgrade driver has also been rewritten to use the Microsoft-supplied WINUSB interface rather than libusb-win32 which is not currently fully supported on Windows 7.

2.11.2 Windows tool Visual Studio projects updated to VS2008 (Reference 12138)

The project files included in the tools directory for the various VisualStudio applications and DLLs have been updated to support VisualStudio2008. They were previously for VisualStudio2005.

2.12 Bug Fixes in EK-LM3S3748 Firmware Package

2.12.1 usb_dev_cserial application not properly echoing data (Reference 12049)

The usb_dev_cserial application did not properly handle echo commands to both serial ports. This caused strings longer than 16 bytes or strings with too many spaces to truncate. Both types of strings are now properly echoed on both serial ports.

2.13 New Features in EK-LM3S6965 Firmware Package

2.13.1 fswrapper module added to ek-lm3s6965 release (Reference 12122)

The file system wrapper module, fswrapper, has been added to the utils directory for the ek-lm3s6965 StellarisWare release. This module allows multiple FAT logical disks and/or internal file system images to be viewed as a single namespace with each appearing under its own top level directory name.

2.14 Bug Fixes in EK-LM3S6965 Firmware Package

2.14.1 RIT display driver used wrong SPI mode (Reference 12217)

The display driver for the RIT 128x96 panel on the evaluation board used the wrong SPI mode. This caused the Stellaris SPI port to transition the data line on the same clock edge that the display controller read the data line. While this managed to work, the correct SPI mode is now used. Additionally, some improvements were made in the handling of the SPI interface, resulting in better performance of the display driver.

2.14.2 Added MPU region to mpu_fault example for bit-banded SRAM (Reference 12235)

A new MPU region has been added to the mpu_fault example to allow read/write access to bit-banded SRAM. This is required by the updated display driver, which uses the bit-banded SRAM region for storing flags.

2.15 New Features in EK-LM3S8962 Firmware Package

2.15.1 fswrapper module added to ek-lm3s8962 release (Reference 12126)

The file system wrapper module, fswrapper, has been added to the utils directory for the ek-lm3s8962 StellarisWare release. This module allows multiple FAT logical disks and/or internal file system images to be viewed as a single namespace with each appearing under its own top level directory name.

2.16 Bug Fixes in EK-LM3S8962 Firmware Package

2.16.1 RIT display driver used wrong SPI mode (Reference 12217)

The display driver for the RIT 128x96 panel on the evaluation board used the wrong SPI mode. This caused the Stellaris SPI port to transition the data line on the same clock edge that the display controller read the data line. While this managed to work, the correct SPI mode is now used. Additionally, some improvements were made in the handling of the SPI interface, resulting in better performance of the display driver.

2.16.2 Added MPU region to mpu_fault example for bit-banded SRAM (Reference 12235)

A new MPU region has been added to the mpu_fault example to allow read/write access to bit-banded SRAM. This is required by the updated display driver, which uses the bit-banded SRAM region for storing flags.

2.17 New Features in EK-LM3S9B90 Firmware Package

2.17.1 USB Drivers Now Support Windows 7 32-bit and 64-bit Versions (Reference 11555)

The SW-USB-windrivers-xxxx.zip USB driver package has been updated to support both 32-bit and 64-bit versions of Windows 7. The package includes 64-bit versions of URLs lmusbdll.dll and lmdfu.dll allowing 64-bit applications to access these interfaces. The Device Firmware Upgrade driver has also been rewritten to use the Microsoft-supplied WINUSB interface rather than libusb-win32 which is not currently fully supported on Windows 7.

2.17.2 Windows tool Visual Studio projects updated to VS2008 (Reference 12138)

The project files included in the tools directory for the various VisualStudio applications and DLLs have been updated to support VisualStudio2008. They were previously for VisualStudio2005.

2.18 Bug Fixes in EK-LM3S9B90 Firmware Package

2.18.1 usb_stick_update would not enumerate drives in all cases. (Reference 12241)

The usb_stick_update only enumerated USB MSC devices if they were present when the application started. The usb_stick_update now properly enumerates devices by forcing the OTG controller to indicate a host connection which allows USB MSC devices to properly enumerate.

2.19 New Features in EK-LM3S9B92 Firmware Package

2.19.1 USB Drivers Now Support Windows 7 32-bit and 64-bit Versions (Reference 11555)

The SW-USB-windrivers-xxxx.zip USB driver package has been updated to support both 32-bit and 64-bit versions of Windows 7. The package includes 64-bit versions of URLs lmusbdll.dll and lmdfu.dll allowing 64-bit applications to access these interfaces. The Device Firmware Upgrade driver has also been rewritten to use the Microsoft-supplied WINUSB interface rather than libusb-win32 which is not currently fully supported on Windows 7.

2.19.2 Windows tool Visual Studio projects updated to VS2008 (Reference 12138)

The project files included in the tools directory for the various VisualStudio applications and DLLs have been updated to support VisualStudio2008. They were previously for VisualStudio2005.

2.20 Bug Fixes in EK-LM3S9B92 Firmware Package

2.20.1 usb_stick_update would not enumerate drives in all cases. (Reference 12241)

The usb_stick_update only enumerated USB MSC devices if they were present when the application started. The usb_stick_update now properly enumerates devices by forcing the OTG controller

to indicate a host connection which allows USB MSC devices to properly enumerate.

2.21 Bug Fixes in RDK-BDC Firmware Package

2.21.1 Fixed API Class table in documentation (Reference 12172)

The API Class table in the documentation did not take into account the fact that 2 and 6 are presently reserved values. This was corrected, and the table now properly documents the values for the API Class field of the protocol.

2.22 Bug Fixes in RDK-BDC24 Firmware Package

2.22.1 Fixed API Class table in documentation (Reference 12172)

The API Class table in the documentation did not take into account the fact that 2 and 6 are presently reserved values. This was corrected, and the table now properly documents the values for the API Class field of the protocol.

2.23 New Features in RDK-IDM-SBC Firmware Package

2.23.1 Windows tool Visual Studio projects updated to VS2008 (Reference 12138)

The project files included in the tools directory for the various VisualStudio applications and DLLs have been updated to support VisualStudio2008. They were previously for VisualStudio2005.

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3.1 New Features in Stellaris Boot Loader

3.1.1 Added support for unlocking forced update pins (Reference 11970)

The boot loader now has the ability to unlock pins that require it (such as PB7/NMI), allowing those pins to be used as forced update pins.

3.1.2 Added optional MOSCFAIL handler to boot loader (Reference 12014)

An optional MOSCFAIL handler has been added to the boot loader to allow proper operation of the MOSC failure detection capability of certain Stellaris microcontrollers. This is enabled via ENABLE_MOSCFAIL_HANDLER in bl_config.h. See the corresponding Stellaris microcontroller data sheet to determine if the device you are using supports MOSC failure detection.

3.2 Bug Fixes in Stellaris USB Library

3.2.1 USB Device CDC had incorrect Configuration Descriptor (Reference 11993)

The USB CDC Serial device class had an incorrect configuration descriptor that was causing some operating systems to not recognize the device. The CDC serial device now properly has two interfaces, one for control and one for data.

3.2.2 The USB Host Pipe Read/Write functions improperly handle STALL (Reference 12024)

The USBHCDPipeWrite() and USBHCDPipeRead() functions both failed to handle stall conditions on endpoints other than zero. This caused the USB library to falsely enter the status phase when it should have terminated the stalled transaction.

3.2.3 USBOTGModeInit() causes ASSERT in debug builds (Reference 12030)

The USBOTGModeInit() function called the USBHostPwrConfig() DriverLib API with bits set that did not have meaning in the USBHostPwrConfig() API which caused it to ASSERT and halt in debug builds. The USBHostPwrConfig() has been replaced with a call to a new API USBHCDPowerConfigSet() which correctly sets the power configuration.

3.3 Bug Fixes in DK-LM3S9B96 Firmware Package

3.3.1 Hang in qs-checkout if run with FS8 daughter containing non-filesystem image (Reference 11976)

The qs-checkout example application could hang if run on a board equipped with the Flash/SRAM/LCD daughter board where the daughter board flash contained data other than a valid file system image. The application now correctly checks for a valid file system image header before trying to access the data.

3.3.2 Internal pull-ups removed from EPI pin configuration (Reference 12019)

The configuration used for each Extended Peripheral Interface pin in set_pinout.c has been changed to remove the internal weak pull-ups since these are not required.

3.3.3 FPGA daughter board initialization problem on power-on-reset (Reference 12034)

In the previous code release, the FPGA/Camera/LCD daughter board would often not initialize correctly after a power-on-reset, requiring the user to press the “Reset” button to restart the application. This problem is due to the fact that the FPGA is configured to pull unconfigured pins up rather than down. The ready signal on PJ6/EPI30 was a low to high transition on one of those pins so this was masked by the fact that the pin was already high. To work around the problem, the initialization code no longer polls PJ6 looking for a rising edge but, instead, delays 600mS after resetting the FPGA and before accessing any of its registers.

3.4 Bug Fixes in EK-LM3S3748 Firmware Package

3.4.1 USB host mass storage applications generate errors on slower devices (Reference 11689)

The applications that use the USB host mass storage class can generate errors if they attempt to access the USB device before it is ready. These applications now ensure that that the USB mass storage device is ready before attempting access by the application.

3.4.2 usb_stick_update application can timeout on retries (Reference 12025)

The usb_stick_update application was only attempting to initialize the USB mass storage device 5 time before giving up. On some larger devices the start up time was much larger than this so the retry count was increased to infinite. The loop is now infinite because the application was previously returning and executing a different infinite loop.

3.5 Bug Fixes in EK-LM3S9B90 Firmware Package

3.5.1 USB host mass storage applications generate errors on slower devices (Reference 11689)

The applications that use the USB host mass storage class can generate errors if they attempt to access the USB device before it is ready. These applications now ensure that that the USB mass storage device is ready before attempting access by the application.

3.5.2 usb_stick_update application can timeout on retries (Reference 12025)

The usb_stick_update application was only attempting to initialize the USB mass storage device 5 time before giving up. On some larger devices the start up time was much larger than this so the retry count was increased to infinite. The loop is now infinite because the application was previously returning and executing a different infinite loop.

3.6 Bug Fixes in EK-LM3S9B92 Firmware Package

3.6.1 USB host mass storage applications generate errors on slower devices (Reference 11689)

The applications that use the USB host mass storage class can generate errors if they attempt to access the USB device before it is ready. These applications now ensure that that the USB mass storage device is ready before attempting access by the application.

3.6.2 usb_stick_update application can timeout on retries (Reference 12025)

The usb_stick_update application was only attempting to initialize the USB mass storage device 5 time before giving up. On some larger devices the start up time was much larger than this so the retry count was increased to infinite. The loop is now infinite because the application was previously returning and executing a different infinite loop.

3.7 Bug Fixes in EK-LM3S9B96 Firmware Package

3.7.1 USB host mass storage applications generate errors on slower devices (Reference 11689)

The applications that use the USB host mass storage class can generate errors if they attempt to access the USB device before it is ready. These applications now ensure that that the USB mass storage device is ready before attempting access by the application.

3.7.2 usb_stick_update application can timeout on retries (Reference 12025)

The usb_stick_update application was only attempting to initialize the USB mass storage device 5 time before giving up. On some larger devices the start up time was much larger than this so the retry count was increased to infinite. The loop is now infinite because the application was previously returning and executing a different infinite loop.

3.8 New Features in RDK-BDC24 Firmware Package

3.8.1 Source Code for bdc-comm now available (Reference 12018)

With this release the source code for bdc-comm is now available in the release package. The source code is located in the tools/bdc-comm directory and includes a Makefile to build the project.

3.9 Bug Fixes in RDK-IDM Firmware Package

3.9.1 Motor speed updates not displayed in bldc_ctrl example (Reference 11826)

The bldc_ctrl example for RDK-IDM previously failed to send a command to the motor to request automatic rotor speed updates. This resulted in the current speed not being displayed even though the motor was running. This has now been corrected.

3.10 Bug Fixes in RDK-IDM-SBC Firmware Package

3.10.1 USB host mass storage applications generate errors on slower devices (Reference 11689)

The applications that use the USB host mass storage class can generate errors if they attempt to access the USB device before it is ready. These applications now ensure that that the USB mass storage device is ready before attempting access by the application.

3.10.2 usb_stick_update application can timeout on retries (Reference 12025)

The usb_stick_update application was only attempting to initialize the USB mass storage device 5 time before giving up. On some larger devices the start up time was much larger than this so the retry count was increased to infinite. The loop is now infinite because the application was previously returning and executing a different infinite loop.

3.11 Bug Fixes in RDK-STEPPER Firmware Package

3.11.1 Fix rare stepper deceleration problem (Reference 11969)

Under a very specific combination of settings, the stepper may not decelerate correctly. This has been fixed.

3.12 New Features in Stellaris Firmware Development Package

3.12.1 Added new set of example applications that are not specific to any board (Reference 12013)

New examples have been added and can be found in the “examples” directory. These examples are not specific to any board and are meant to demonstrate one specific feature or mode of a peripheral. There is also a new document summarizing these examples in the “docs” directory.

3.13 Bug Fixes in Stellaris Firmware Development Package

3.13.1 Corrected uVision 3 project files (Reference 11990)

The Keil uVision 3 project files failed to list the part number, memory size, flash programming algorithm, and so on. These have been corrected.

4 Release Notes for StellarisWare Revision 5879 (April 14, 2010)

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4.1 New Features in Stellaris USB Library

4.1.1 Added USB Composite Device support (Reference 11721)

This version of the Stellaris USB library adds support for using multiple USB device classes in a single composite device. The USB library documentation includes more details covering how to use this new feature.

4.2 Bug Fixes in Stellaris USB Library

4.2.1 USBHCDPipeRead() returns 0 bytes in some cases (Reference 11880)

In cases where USBHCDPipeRead() is attempting to read packets of less than 64 bytes, the function will return that it has read 0 bytes. The USBHCDPipeRead() function will now properly set the number of bytes read instead of always returning zero bytes for packets that are less than 64 bytes.

4.3 Bug Fixes in DK-LM3S9B96 Firmware Package

4.3.1 Display initialization parameters updated (Reference 11884)

The initialization parameters for the Kitronix 3.5" display on the IDM-SBC, IDM-L35 and DK-LM3S9B96 boards have been updated to reflect recommendations received from the display manufacturer.

4.3.2 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.3.3 USB CDC serial device examples incorrectly responding to Get Line Coding (Reference 11767)

When sending a response to the Get Line Coding request, the GetLineCoding() function was returning data directly to endpoint 0 instead of allowing the USB library to send the data. This could generate multiple IN data packets to the host and cause the library to get out of sync.

4.4 Bug Fixes in EK-LM3S3748 Firmware Package

4.4.1 Missing files added to tools/lmscope (Reference 11869)

Four source files required to build the lmscope Windows example application were inadvertently omitted in previous StellarisWare releases. These files, lmusbwrap.cpp, lmusbwrap.h, TI_Symbol.bmp and TILogo.bmp, have now been added to the release.

4.4.2 USB CDC serial device examples incorrectly responding to Get Line Coding (Reference 11767)

When sending a response to the Get Line Coding request, the GetLineCoding() function was returning data directly to endpoint 0 instead of allowing the USB library to send the data. This could generate multiple IN data packets to the host and cause the library to get out of sync.

4.5 New Features in EK-LM3S6965 Firmware Package

4.5.1 boot_demo_eth example application added. (Reference 11910)

The boot_demo_eth example application, illustrating the use of the swupdate module, has been added to ek-lm3s9b96 and ek-lm3s6965 releases of StellarisWare. The swupdate module is used in conjunction with the Ethernet boot loader to signal to the application when a firmware update request is received from LMFlash.

4.6 Bug Fixes in EK-LM3S6965 Firmware Package

4.6.1 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.6.2 HTTP/SSI Performance Improvement (Reference 11935)

A misplaced “return” instruction inside the SSI handling state machine of the lwIP HTTP server caused each SSI insert to be returned to the client in a single TCP packet, reducing performance. Removing this allows the server to buffer up as many SSI inserts as can be handled in the available TCP send buffer before sending them as a single packet, thus improving performance when serving pages containing many SSI tags.

4.7 Bug Fixes in EK-LM3S8962 Firmware Package

4.7.1 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.7.2 HTTP/SSI Performance Improvement (Reference 11935)

A misplaced “return” instruction inside the SSI handling state machine of the lwIP HTTP server caused each SSI insert to be returned to the client in a single TCP packet, reducing performance. Removing this allows the server to buffer up as many SSI inserts as can be handled in the available TCP send buffer before sending them as a single packet, thus improving performance when serving pages containing many SSI tags.

4.8 Bug Fixes in EK-LM3S9B90 Firmware Package

4.8.1 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.8.2 USB CDC serial device examples incorrectly responding to Get Line Coding (Reference 11767)

When sending a response to the Get Line Coding request, the GetLineCoding() function was returning data directly to endpoint 0 instead of allowing the USB library to send the data. This could generate multiple IN data packets to the host and cause the library to get out of sync.

4.8.3 HTTP/SSI Performance Improvement (Reference 11935)

A misplaced “return” instruction inside the SSI handling state machine of the lwIP HTTP server caused each SSI insert to be returned to the client in a single TCP packet, reducing performance. Removing this allows the server to buffer up as many SSI inserts as can be handled in the available TCP send buffer before sending them as a single packet, thus improving performance when serving pages containing many SSI tags.

4.9 Bug Fixes in EK-LM3S9B92 Firmware Package

4.9.1 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of

time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.9.2 USB CDC serial device examples incorrectly responding to Get Line Coding (Reference 11767)

When sending a response to the Get Line Coding request, the GetLineCoding() function was returning data directly to endpoint 0 instead of allowing the USB library to send the data. This could generate multiple IN data packets to the host and cause the library to get out of sync.

4.10 Bug Fixes in EK-LM3S9B96 Firmware Package

4.10.1 HTTP/SSI Performance Improvement (Reference 11935)

A misplaced “return” instruction inside the SSI handling state machine of the lwIP HTTP server caused each SSI insert to be returned to the client in a single TCP packet, reducing performance. Removing this allows the server to buffer up as many SSI inserts as can be handled in the available TCP send buffer before sending them as a single packet, thus improving performance when serving pages containing many SSI tags.

4.11 Bug Fixes in RDK-BLDC Firmware Package

4.11.1 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.12 Bug Fixes in RDK-IDM Firmware Package

4.12.1 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of

time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.12.2 HTTP/SSI Performance Improvement (Reference 11935)

A misplaced “return” instruction inside the SSI handling state machine of the lwIP HTTP server caused each SSI insert to be returned to the client in a single TCP packet, reducing performance. Removing this allows the server to buffer up as many SSI inserts as can be handled in the available TCP send buffer before sending them as a single packet, thus improving performance when serving pages containing many SSI tags.

4.13 Bug Fixes in RDK-IDM-L35 Firmware Package

4.13.1 Display initialization parameters updated (Reference 11884)

The initialization parameters for the Kitronix 3.5" display on the IDM-SBC, IDM-L35 and DK-LM3S9B96 boards have been updated to reflect recommendations received from the display manufacturer.

4.14 Bug Fixes in RDK-IDM-SBC Firmware Package

4.14.1 Fix LocatorAppTitleSet strings (Reference 11829)

The application string for a couple of the RDK-IDM-SBC applications had a cut-and-paste error. These strings were fixed to reflect the correct application name.

4.14.2 Display initialization parameters updated (Reference 11884)

The initialization parameters for the Kitronix 3.5" display on the IDM-SBC, IDM-L35 and DK-LM3S9B96 boards have been updated to reflect recommendations received from the display manufacturer.

4.14.3 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for

each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.14.4 HTTP/SSI Performance Improvement (Reference 11935)

A misplaced “return” instruction inside the SSI handling state machine of the lwIP HTTP server caused each SSI insert to be returned to the client in a single TCP packet, reducing performance. Removing this allows the server to buffer up as many SSI inserts as can be handled in the available TCP send buffer before sending them as a single packet, thus improving performance when serving pages containing many SSI tags.

4.15 Bug Fixes in RDK-S2E Firmware Package

4.15.1 Shorten time to select link-local address in lwIP applications (Reference 11908)

In lwIP 1.3.1, the time delay between DHCP discover messages was changed from a linear delay to an exponential delay (to conform with the relevant standards). This results in a longer amount of time (around 4 minutes) before a link-local address is chosen instead of a DHCP-assigned address since this occurs after a set number of DHCP discover messages. Therefore, the lwipopts.h for each lwIP application now sets LWIP_DHCP_AUTOIP_COOP_TRIES to 5 (instead of its default 9) so that the link local address is chosen after around 40 seconds.

4.15.2 HTTP/SSI Performance Improvement (Reference 11935)

A misplaced “return” instruction inside the SSI handling state machine of the lwIP HTTP server caused each SSI insert to be returned to the client in a single TCP packet, reducing performance. Removing this allows the server to buffer up as many SSI inserts as can be handled in the available TCP send buffer before sending them as a single packet, thus improving performance when serving pages containing many SSI tags.

4.16 Bug Fixes in Stellaris Firmware Development Package

4.16.1 Clarify parameter block size requirements in FlashPBInit() (Reference 11888)

The flash parameter block size requirements were clarified to indicate that parameter flash storage size divided by the parameter block size (in other words, the number of parameter block instances that can be stored in flash) must be less than or equal to 128.

5 Release Notes for StellarisWare Revision 5821 (March 29, 2010)

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5.1 New Features in Stellaris Peripheral Driver Library

5.1.1 Added TimerPrescaleMatchSet and TimerPrescaleMatchGet (Reference 11711)

The TimerPrescaleMatchSet and TimerPrescaleMatchGet APIs were added to the timer driver to allow the prescale match functionality that is available in newer parts to be accessed.

5.1.2 Added new part specific headers (Reference 11755)

This release added new part specific headers for LM3S3634, LM3S1P51, LM3S1R21, and LM3S1651.

5.1.3 Added ADCPhaseDelaySet and ADCPhaseDelayGet to ADC driver (Reference 11770)

The ADCPhaseDelaySet and ADCPhaseDelayGet APIs were added to allow the phase delay between multiple ADC modules to be controlled. With phase delay and more than one ADC module sampling the same input, each can sample at different times, effectively doubling the sampling frequency of the analog input.

5.1.4 Add ability to synchronize the ADC processor trigger (Reference 11769)

The ADCProcessorTrigger API was extended to allow a processor trigger to be sent to multiple ADC modules in a synchronous manner. Each module is individually triggered, which each being

told to wait to handle the trigger until a synchronization signal.

5.1.5 Added GPIOPinTypeEPI to GPIO driver (Reference 9972)

The GPIOPinTypeEPI API has been added to the GPIO driver. It provides a typical configuration of a pin for use by the EPI module.

5.1.6 Added ROM call definitions for Tempest-class revision C1 (Reference 11783)

Definitions were added to driverlib/rom.h and driverlib/rom_map.h to allow functions in ROM to be called on revision C1 of the Tempest-class parts.

5.2 Bug Fixes in Stellaris Peripheral Driver Library

5.2.1 FlashIntGetStatus() renamed FlashIntStatus() (Reference 11698)

To conform to the naming convention used in all other DriverLib peripheral APIs, the function FlashIntGetStatus() has been renamed FlashIntStatus(). A definition has been added to flash.h to ensure that code using the previous function name will still build but please note that the previous function name is deprecated and may be removed at some point in the future. Two new defines, FLASH_INT_PROGRAM and FLASH_INT_ERASE, have also been added to flash.h for use by the various interrupt-related functions and the documentation for these functions updated to show the correct valid parameter values.

5.2.2 SysCtlADCSpeedSet now sets the speed of ADC1 (Reference 11768)

Previous, SysCtlADCSpeedSet would set the sample rate of ADC0 to the requested speed and unconditionally set ADC1 to 125 ksps. Now, it will set both ADCs to the requested speed.

5.3 Bug Fixes in Stellaris Utility Library

5.3.1 Add missing timer callbacks to lwiplib (Reference 11696)

Support for timer callbacks for IGMP, IP Reassembly, and DNS were missing from the lwiplib.c timer service routine. Add support for these timer callbacks to the timer service routine for the case were no RTOS is being supported. If an RTOS is being used, these routines are already supported properly in the TCP thread.

5.4 Bug Fixes in DK-LM3S9B96 Firmware Package

5.4.1 In usb_stick_demo, delay call to usb_stick_update until button is released (Reference 11692)

Previously, usb_stick_demo would call usb_stick_update as soon as the push button is pressed. It is possible for the update to occur prior to the button being released, in which case the update will occur again since usb_stick_update uses the pressed state of the button as a signal to update instead of calling the application. Now, usb_stick_update is not called until the button is pressed and then released, removing this possibility.

5.4.2 Example boot_eth_ext failed to write flash. (Reference 11798)

Due to a recent change which prevented the Ethernet boot loader from writing past the end of internal flash, the external flash boot loader example in release 5727 was unable to write to the EPI address space at 0x60000000. This has been corrected and the example works correctly once again.

5.4.3 Web server performance improved in safertos_demo. (Reference 11685)

Resource allocation for lwIP was increased to improve the performance of the web server in the safertos_demo example application. Prior to this change, load times for the lwIP statistics page served by the application were extremely long when using some browsers including Firefox for Windows.

5.4.4 USB_VID_LUMINARY replaced with USB_VID_STELLARIS (Reference 11808)

The label USB_VID_LUMINARY, defining the USB vendor ID for Luminary Micro, has been replaced by USB_VID_STELLARIS. The new label is defined to the same value as the one it replaces, namely 0x1cbe. The previous label remains defined in header file usb-ids.h but is marked as deprecated.

5.5 Bug Fixes in EK-LM3S3748 Firmware Package

5.5.1 In usb_stick_demo, delay call to usb_stick_update until button is released (Reference 11692)

Previously, usb_stick_demo would call usb_stick_update as soon as the push button is pressed. It is possible for the update to occur prior to the button being released, in which case the update

will occur again since usb_stick_update uses the pressed state of the button as a signal to update instead of calling the application. Now, usb_stick_update is not called until the button is pressed and then released, removing this possibility.

5.5.2 USB_VID_LUMINARY replaced with USB_VID_STELLARIS (Reference 11808)

The label USB_VID_LUMINARY, defining the USB vendor ID for Luminary Micro, has been replaced by USB_VID_STELLARIS. The new label is defined to the same value as the one it replaces, namely 0x1cbe. The previous label remains defined in header file usb-ids.h but is marked as deprecated.

5.6 Bug Fixes in EK-LM3S9B90 Firmware Package

5.6.1 In usb_stick_demo, delay call to usb_stick_update until button is released (Reference 11692)

Previously, usb_stick_demo would call usb_stick_update as soon as the push button is pressed. It is possible for the update to occur prior to the button being released, in which case the update will occur again since usb_stick_update uses the pressed state of the button as a signal to update instead of calling the application. Now, usb_stick_update is not called until the button is pressed and then released, removing this possibility.

5.6.2 Added explicit GPIOPinConfigure calls to examples (Reference 11733)

Explicit GPIOPinConfigure calls for the UART pins were added to all the example application to help clarify the requirement to use this call to configure the pins. While not required for UART0 (since it is the default configuration of those pins), it would be required if the code was used as a basis for something that uses UART1.

5.6.3 USB_VID_LUMINARY replaced with USB_VID_STELLARIS (Reference 11808)

The label USB_VID_LUMINARY, defining the USB vendor ID for Luminary Micro, has been replaced by USB_VID_STELLARIS. The new label is defined to the same value as the one it replaces, namely 0x1cbe. The previous label remains defined in header file usb-ids.h but is marked as deprecated.

5.7 Bug Fixes in EK-LM3S9B92 Firmware Package

5.7.1 In usb_stick_demo, delay call to usb_stick_update until button is released (Reference 11692)

Previously, usb_stick_demo would call usb_stick_update as soon as the push button is pressed. It is possible for the update to occur prior to the button being released, in which case the update will occur again since usb_stick_update uses the pressed state of the button as a signal to update instead of calling the application. Now, usb_stick_update is not called until the button is pressed and then released, removing this possibility.

5.7.2 Added explicit GPIOPinConfigure calls to examples (Reference 11733)

Explicit GPIOPinConfigure calls for the UART pins were added to all the example application to help clarify the requirement to use this call to configure the pins. While not required for UART0 (since it is the default configuration of those pins), it would be required if the code was used as a basis for something that uses UART1.

5.7.3 USB_VID_LUMINARY replaced with USB_VID_STELLARIS (Reference 11808)

The label USB_VID_LUMINARY, defining the USB vendor ID for Luminary Micro, has been replaced by USB_VID_STELLARIS. The new label is defined to the same value as the one it replaces, namely 0x1cbe. The previous label remains defined in header file usb-ids.h but is marked as deprecated.

5.8 Bug Fixes in RDK-BDC Firmware Package

5.8.1 Corrected speed sensing when driving in the negative direction (Reference 11752)

When driving the motor in the negative direction, the sensed motor speed would occasionally appear to the PID loop as a positive speed despite the fact that the motor is turning in the negative direction. This has been corrected, providing better PID loop performance in the reverse direction (it is now symmetrical to the forward direction).

5.8.2 Prevent roll over in the PID integrator (Reference 11753)

When presented with a large error term, it was possible for the integrator in the PID loop to roll over (from positive to negative or negative to positive), causing the PID loop to fail to control the motor as expected. This roll over is now prevented and the PID loop behaves as expected.

5.8.3 Corrected current sensing when driving in the negative direction (Reference 11797)

When driving the motor in the negative direction, the sensed motor current would occasionally appear to the PID loop as a positive current despite the fact that the motor is turning in the negative direction. This has been corrected, providing better PID loop performance in the reverse direction (it is now symmetrical to the forward direction).

5.9 Bug Fixes in RDK-BDC24 Firmware Package

5.9.1 Corrected speed sensing when driving in the negative direction (Reference 11752)

When driving the motor in the negative direction, the sensed motor speed would occasionally appear to the PID loop as a positive speed despite the fact that the motor is turning in the negative direction. This has been corrected, providing better PID loop performance in the reverse direction (it is now symmetrical to the forward direction).

5.9.2 Prevent roll over in the PID integrator (Reference 11753)

When presented with a large error term, it was possible for the integrator in the PID loop to roll over (from positive to negative or negative to positive), causing the PID loop to fail to control the motor as expected. This roll over is now prevented and the PID loop behaves as expected.

5.9.3 Corrected current sensing when driving in the negative direction (Reference 11797)

When driving the motor in the negative direction, the sensed motor current would occasionally appear to the PID loop as a positive current despite the fact that the motor is turning in the negative direction. This has been corrected, providing better PID loop performance in the reverse direction (it is now symmetrical to the forward direction).

5.10 Bug Fixes in RDK-IDM Firmware Package

5.10.1 bldc_ctrl example rebranded (Reference 11713)

The bldc_ctrl example has been updated to show Texas Instruments branding in place of the previous Luminary Micro text and logo.

5.11 Bug Fixes in Stellaris Firmware Development Package

5.11.1 Modified CCS linker command file to ensure proper location of vtable (Reference 11710)

If the application used dynamic interrupt registration (through an IntRegister function), sometimes the vector table in RAM (vtable) would not be located correctly. This only affected CCS projects and only if the vtable was used which is not usual. This problem has been fixed by explicitly setting the location of vtable in the linker command file.

6 Release Notes for StellarisWare Revision 5727 (March 2, 2010)

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6.1 Bug Fixes in Stellaris Boot Loader

6.1.1 Ethernet boot loader fails to ACK last packet in some cases (Reference 11552)

When the last portion of a file being transferred via TFTP is contained in a full data packet of 512 bytes, an additional empty data packet is subsequently sent to indicate that the transfer has completed. When this occurs at the end of Flash, the empty packet was incorrectly responded to with an error packet instead of an ACK packet. It will now be ACKed, allowing the TFTP server (such as eflash) to properly detect the end of the transfer.

6.2 New Features in Stellaris Peripheral Driver Library

6.2.1 New automatic and manual USB power control (Reference 11469)

The USB DriverLib power configuration function was not allowing the Stellaris USB controller to automatically power VBUS in OTG mode and was forcing the application or USB library to use a manual power method. Since the manual power method is still desirable and allows an ap-

plication to use it's own method to power VBUS, this mode remains and new mode was added to allow the USB controller full control over powering VBUS. This change added the new USBHostPwrConfig() power configuration function that replaces the old USBHostPwrFaultConfig() function. The old USB_HOST_PWREN_* values and the USBHostPwrFaultConfig() function have been deprecated and new values have been added to use with the new USBHostPwrConfig() function. The USB_HOST_PWREN_MAN_LOW and USB_HOST_PWREN_MAN_HIGH allow the application to control when power is applied to VBUS and USB_HOST_PWREN_AUTOLOW and USB_HOST_PWREN_AUTOHIGH will allow the USB OTG controller to have full control over powering to VBUS.

6.2.2 Add ADC functions to support use of an external reference (Reference 11494)

Added two new ADC functions, ADCReferenceSet() and ADCReferenceGet(). These can be used to select internal or external reference for those parts that support an external reference.

6.3 Bug Fixes in Stellaris Peripheral Driver Library

6.3.1 uDMA function uDMAChannelSizeGet() was returning the wrong value for completed transfers (Reference 11632)

When all of the items had been transferred and the transfer was complete, this function was returning a 1 when it should have returned a 0. This has now been fixed. For all other conditions, this function was returning the correct value.

6.4 New Features in Stellaris USB Library

6.4.1 Added the ability to use the USB VBUS filter (Reference 11339)

Some Stellaris microcontrollers with USB OTG functionality have the ability to ignore short dips in VBUS. These momentary dips are usually caused by inrush current to a device and can drop VBUS below VBUS valid. When VBUS drops below VBUS valid level the USB OTG controller will exit host mode and drop VBUS to below session valid level and attempt to power the device again. The VBUS filter is enabled by calling the USBHostPwrConfig() function with the USB_HOST_PWREN_FILTER bit set.

6.4.2 Added Isochronous Host controller support to USB Library (Reference 11528)

The USB library did not have support for using the Host controller with Isochronous devices. This update adds support for USB Isochronous Audio devices and provides an example of how to enumerate other USB Isochronous devices and interact with Isochronous IN and OUT endpoints. See

the USB library documentation for more information on using the USB Host Isochronous audio class.

6.4.3 Added new function to allocate FIFO to an endpoint (Reference 11557)

The USBHCDPipeAlloc() was the only method used to allocate USB FIFO memory to endpoints and used a fixed allocation of 64 bytes per endpoint. To work with endpoints that have larger FIFO requirements, like Isochronous endpoints, a new USBHCDPipeAllocSize() function was added that adds a size parameter to attempt to allocate memory to an endpoint. This allows Isochronous endpoints to allocate up to 1024 bytes to a single endpoint while still allowing Bulk, Interrupt and Control endpoints to use the standard 64 byte FIFOs.

6.5 Bug Fixes in Stellaris USB Library

6.5.1 USBlib overwrites power configuration in OTG mode (Reference 11338)

When using OTG mode, usblib was using a fixed setting for the power control pins and ignoring the application supplied settings. OTG mode now correctly uses the settings that the application provides to usblib.

6.5.2 USB Control requests can hang on error (Reference 11326)

If an error state occurs on USB control request to endpoint 0, it could cause the USB library to hang while attempting to enumerate a USB device. The control requests to endpoint 0 will now terminate in the event of an error and allow the USB library and an application to respond to the error. This issue could affect any USB device during enumeration, however it was having a more obvious effect on some USB Mass Storage devices.

6.5.3 Host enumeration was incorrectly requesting a zero byte packet (Reference 11517)

The USB library host enumeration code was incorrectly requesting an extra zero length packet when reading descriptors from a USB device. This could cause the device to Stall the transaction and the USB library would then fail to enumerate the device. This only happened when a USB device had any 64 byte aligned descriptors.

6.6 New Features in Stellaris Utility Library

6.6.1 Added features to bdc-comm (Reference 11321)

Several new features have been added to the bdc-comm GUI. There is now a mechanism for recovering a MDL-BDC or MDL-BDC24 that has had the incorrect firmware programmed into it (accessed via the File->Recover Device menu item). It is now possible to assign device IDs to a MDL-BDC or MDL-BDC24 even if bdc-comm can not find any devices on the network (they may be there without an assigned ID and therefore do not enumerate). The Help->About menu item was added, which brings up a dialog that shows the version of the bdc-comm applcation. The numeric entry fields within the GUI have been modified to behave in the expected manner (click and drag will now select portions of the value instead of changing the value). And the firmware filename field in the firmware update dialog is now pre-populated with the previous firmware filename so that it can be used multiple times to update more than one MDL-BDC or MDL-BDC24.

6.7 Bug Fixes in Stellaris Utility Library

6.7.1 Correct leap day handling in ulocaltime (Reference 11049)

Leap days were not properly handled in ulocaltime, causing it to incorrectly report Feb 29 of a leap year as Mar 1, and Mar 1 of a leap year as Mar 2 (with all other days being reported correctly). It now properly handles leap days.

6.8 New Features in Third Party Packages

6.8.1 Upgraded to FLTK 1.1.10 (Reference 11551)

The FLTK host toolkit was upgraded from version 1.1.9 to 1.1.10, and the applications which use it updated as well. This avoid the compile errors seen with some flavors of Linux when using 1.1.9 (therefore allowing the host utilities that use FLTK to build under Linux).

6.9 Bug Fixes in Third Party Packages

6.9.1 Explicit casts added to lwIP TCP_SEQ_LT/LTE/GT/GTE macros (Reference 11600)

Explicit casts have been added to the parameters passed to the TCP_SEQ_LT, TCP_SEQ_LTE, TCP_SEQ_GT and TCP_SEQ_GTE macros in lwIP header file tcp.h to ensure that these versions behave the same way as the original macros regardless of whether they are passed signed or unsigned values. These macros were replaced in the previous release to work around an IAR 5.41 compiler bug. The edit will be removed once a toolchain fix is available.

6.10 Bug Fixes in DK-LM3S9B96 Firmware Package

6.10.1 Rework of lwIP HTTPD debug message handling (Reference 11480)

Debug message handling in the TI-modified lwIP HTTPD server has been reworked such that it follows the same convention as other lwIP modules. Calls to the UARTprintf() function have been replaced with the LWIP_DEBUGF macro. This change has no impact on non-debug builds of the server.

6.10.2 Explicit casts added to lwIP TCP_SEQ_LT/LTE/GT/GTE macros (Reference 11600)

Explicit casts have been added to the parameters passed to the TCP_SEQ_LT, TCP_SEQ_LTE, TCP_SEQ_GT and TCP_SEQ_GTE macros in lwIP header file tcp.h to ensure that these versions behave the same way as the original macros regardless of whether they are passed signed or unsigned values. These macros were replaced in the previous release to work around an IAR 5.41 compiler bug. The edit will be removed once a toolchain fix is available.

6.11 Bug Fixes in EK-LM3S3748 Firmware Package

6.11.1 Modify udma_demo example app to work around known problem with DMA channel (Reference 11640)

There is a chip erratum for this part related to incorrect operation of the DMA channel dedicated to software initiated transfers. The example app was modified to implement the simple workaround.

6.12 Bug Fixes in EK-LM3S6965 Firmware Package

6.12.1 Rework of lwIP HTTPD debug message handling (Reference 11480)

Debug message handling in the TI-modified lwIP HTTPD server has been reworked such that it follows the same convention as other lwIP modules. Calls to the UARTprintf() function have been replaced with the LWIP_DEBUGF macro. This change has no impact on non-debug builds of the server.

6.12.2 Explicit casts added to lwIP TCP_SEQ_LT/LTE/GT/GTE macros (Reference 11600)

Explicit casts have been added to the parameters passed to the TCP_SEQ_LT, TCP_SEQ_LTE, TCP_SEQ_GT and TCP_SEQ_GTE macros in lwIP header file tcp.h to ensure that these versions

behave the same way as the original macros regardless of whether they are passed signed or unsigned values. These macros were replaced in the previous release to work around an IAR 5.41 compiler bug. The edit will be removed once a toolchain fix is available.

6.13 Bug Fixes in EK-LM3S8962 Firmware Package

6.13.1 Rework of lwIP HTTPD debug message handling (Reference 11480)

Debug message handling in the TI-modified lwIP HTTPD server has been reworked such that it follows the same convention as other lwIP modules. Calls to the UARTprintf() function have been replaced with the LWIP_DEBUGF macro. This change has no impact on non-debug builds of the server.

6.13.2 Explicit casts added to lwIP TCP_SEQ_LT/LTE/GT/GTE macros (Reference 11600)

Explicit casts have been added to the parameters passed to the TCP_SEQ_LT, TCP_SEQ_LTE, TCP_SEQ_GT and TCP_SEQ_GTE macros in lwIP header file tcp.h to ensure that these versions behave the same way as the original macros regardless of whether they are passed signed or unsigned values. These macros were replaced in the previous release to work around an IAR 5.41 compiler bug. The edit will be removed once a toolchain fix is available.

6.14 Bug Fixes in EK-LM3S9B90 Firmware Package

6.14.1 Remove udma_timer_ccp example from EK-LM3S9B90 board because the LM3S9B90 does not have PWM (Reference 11476)

The udma_timer_ccp example uses PWM as a signal source for the timer edge-timer feature. The LM3S9B90 chip does not have PWM

6.14.2 Rework of lwIP HTTPD debug message handling (Reference 11480)

Debug message handling in the TI-modified lwIP HTTPD server has been reworked such that it follows the same convention as other lwIP modules. Calls to the UARTprintf() function have been replaced with the LWIP_DEBUGF macro. This change has no impact on non-debug builds of the server.

6.14.3 Explicit casts added to lwIP TCP_SEQ_LT/LTE/GT/GTE macros (Reference 11600)

Explicit casts have been added to the parameters passed to the TCP_SEQ_LT, TCP_SEQ_LTE, TCP_SEQ_GT and TCP_SEQ_GTE macros in lwIP header file tcp.h to ensure that these versions behave the same way as the original macros regardless of whether they are passed signed or unsigned values. These macros were replaced in the previous release to work around an IAR 5.41 compiler bug. The edit will be removed once a toolchain fix is available.

6.15 Bug Fixes in EK-LM3S9B92 Firmware Package

6.15.1 Rework of lwIP HTTPD debug message handling (Reference 11480)

Debug message handling in the TI-modified lwIP HTTPD server has been reworked such that it follows the same convention as other lwIP modules. Calls to the UARTprintf() function have been replaced with the LWIP_DEBUGF macro. This change has no impact on non-debug builds of the server.

6.15.2 Changed timer configuration in udma_timer_ccp example (Reference 11591)

The timer load value corrected and an unneeded function call to set the prescaler was removed.

6.16 Bug Fixes in RDK-BDC Firmware Package

6.16.1 MDL-BDC sometimes misread the power-on state of the push button (Reference 11586)

In certain situations, the MDL-BDC would misread the power-on state of the push button, causing it to reset its stored settings (servo input calibration and CAN device ID assignment). A delay was added between the enable and the read of the GPIO to allow the RC time constant of the push button circuit to pass so that the button state is properly read.

6.17 Bug Fixes in RDK-IDM Firmware Package

6.17.1 Rework of lwIP HTTPD debug message handling (Reference 11480)

Debug message handling in the TI-modified lwIP HTTPD server has been reworked such that it follows the same convention as other lwIP modules. Calls to the UARTprintf() function have been

replaced with the LWIP_DEBUGF macro. This change has no impact on non-debug builds of the server.

6.17.2 Explicit casts added to lwIP TCP_SEQ_LT/LTE/GT/GTE macros (Reference 11600)

Explicit casts have been added to the parameters passed to the TCP_SEQ_LT, TCP_SEQ_LTE, TCP_SEQ_GT and TCP_SEQ_GTE macros in lwIP header file tcp.h to ensure that these versions behave the same way as the original macros regardless of whether they are passed signed or unsigned values. These macros were replaced in the previous release to work around an IAR 5.41 compiler bug. The edit will be removed once a toolchain fix is available.

6.18 Bug Fixes in RDK-IDM-SBC Firmware Package

6.18.1 Rework of lwIP HTTPD debug message handling (Reference 11480)

Debug message handling in the TI-modified lwIP HTTPD server has been reworked such that it follows the same convention as other lwIP modules. Calls to the UARTprintf() function have been replaced with the LWIP_DEBUGF macro. This change has no impact on non-debug builds of the server.

6.18.2 Removed non-functional link from idm-checkout web site (Reference 11658)

The web site served by the idm-checkout application previously included an erroneous link indicating that HTML could be served from an attached USB flash stick. This particular example does not include support for USB flash sticks so this link has been removed.

6.19 Bug Fixes in RDK-S2E Firmware Package

6.19.1 Rework of lwIP HTTPD debug message handling (Reference 11480)

Debug message handling in the TI-modified lwIP HTTPD server has been reworked such that it follows the same convention as other lwIP modules. Calls to the UARTprintf() function have been replaced with the LWIP_DEBUGF macro. This change has no impact on non-debug builds of the server.

6.19.2 Explicit casts added to lwIP TCP_SEQ_LT/LTE/GT/GTE macros (Reference 11600)

Explicit casts have been added to the parameters passed to the TCP_SEQ_LT, TCP_SEQ_LTE, TCP_SEQ_GT and TCP_SEQ_GTE macros in lwIP header file tcp.h to ensure that these versions behave the same way as the original macros regardless of whether they are passed signed or unsigned values. These macros were replaced in the previous release to work around an IAR 5.41 compiler bug. The edit will be removed once a toolchain fix is available.

6.20 New Features in Stellaris Firmware Development Package

6.20.1 Added project files for Keil uVision 4 (Reference 11578)

Project files are now provided for building the various components of StellarisWare using Keil uVision 4. The uVision 3 project files are still provided to ease the transition to Keil's new tools.

6.21 Bug Fixes in Stellaris Firmware Development Package

6.21.1 Projects for Code Red's Red Suite were erroneously including a subdirectory for Code Composer Studio (Reference 11465)

When a project is imported into a workspace using Code Red's red_suite, a subdirectory named "ccs" was also imported by mistake. This subdirectory contains files relevant to TI's Code Composer Studio and should not be part of a red_suite project. This change fixes the problem. The mistaken import of "ccs", while a nuisance, did not actually cause any problems for the red_suite project.

6.21.2 Some applications linked to wrong address with Sourcery G++ (Reference 11624)

For applications that provided a customized lm3sNNNN-rom.ld linker script (to modify the memory descriptor so that the application is linked to an address other than 0), that linker script was not being included into the Sourcery G++ project file. It was therefore using the default linker script and linking the application to 0. The customized linker script is now included as part of the project, resulting in the application being linked correctly.

7 Release Notes for StellarisWare Revision 5604 (January 19, 2010)

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7.1 New Features in Stellaris Peripheral Driver Library

7.1.1 REVISION_IS_C0 macro added to hw_types.h (Reference 11274)

File hw_types.h contains macros to check for particular silicon major and minor revisions. Although macros already existed for revisions C1 and C2, no definition was included for C0 so this has been added for completeness.

7.2 Bug Fixes in Stellaris Host Tools

7.2.1 Improvements to bdc-comm (Reference 11260)

Functional and cosmetic improvements have been made to the bdc-comm application. In some cases, the previous version would fail to send commands to the RDK-BDC24 in response to GUI actions. In other cases, the value set via the GUI would read back as off by one (for example, setting the I coefficient to 0.005 would be read back and displayed as 0.004). These issues have been corrected.

7.3 Bug Fixes in Stellaris USB Library

7.3.1 USB Host: Polling interval from devices was ignored. (Reference 11068)

The USB host enumeration library was not using the polling interval specified by USB devices that were connected. This caused HID devices to poll at a 1ms rate instead of what the HID device specified in it's descriptor. The polling interval is no longer ignored by the USBHCDPipeConfig() function and will now set the polling interval for endpoints correctly.

7.4 New Features in EK-LM3S1968 Firmware Package

7.4.1 Changed logo in “graphics” example application (Reference 11220)

The logo displayed by the “graphics” example application has been replaced. The application has also been updated such that it includes an unaltered 4bpp Windows bitmap and parses the dimensions out of this. In previous versions, the image was a Windows bitmap with the header information removed and dimensions hard-coded. This change is intended to make it easier for users to change the graphic.

7.5 New Features in EK-LM3S2965 Firmware Package

7.5.1 Changed logo in “graphics” example application (Reference 11220)

The logo displayed by the “graphics” example application has been replaced. The application has also been updated such that it includes an unaltered 4bpp Windows bitmap and parses the dimensions out of this. In previous versions, the image was a Windows bitmap with the header information removed and dimensions hard-coded. This change is intended to make it easier for users to change the graphic.

7.6 New Features in EK-LM3S6965 Firmware Package

7.6.1 Changed logo in “graphics” example application (Reference 11220)

The logo displayed by the “graphics” example application has been replaced. The application has also been updated such that it includes an unaltered 4bpp Windows bitmap and parses the dimensions out of this. In previous versions, the image was a Windows bitmap with the header information removed and dimensions hard-coded. This change is intended to make it easier for users to change the graphic.

7.7 New Features in EK-LM3S8962 Firmware Package

7.7.1 Changed logo in “graphics” example application (Reference 11220)

The logo displayed by the “graphics” example application has been replaced. The application has also been updated such that it includes an unaltered 4bpp Windows bitmap and parses the dimensions out of this. In previous versions, the image was a Windows bitmap with the header information removed and dimensions hard-coded. This change is intended to make it easier for users to change the graphic.

7.8 New Features in RDK-BDC Firmware Package

7.8.1 Added a command to return the control mode. (Reference 11262)

A new status command was added to request the control mode for a motor controller. This command returns the last control mode that the controller was set to as an 8-bit value indicating Voltage, Current, Position, or Speed mode.

7.9 New Features in RDK-BDC24 Firmware Package

7.9.1 Added a command to return the control mode. (Reference 11262)

A new status command was added to request the control mode for a motor controller. This command returns the last control mode that the controller was set to as an 8-bit value indicating Voltage, Current, Position, or Speed mode.

8 Release Notes for StellarisWare Revision 5570 (January 8, 2010)

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8.1 New Features in Stellaris Peripheral Driver Library

8.1.1 Added header files for new Stellaris parts (Reference 11183)

Added part-specific header files for the recently introduced LM3S1811, LM3S1816, LM3S1J11, LM3S1J16, LM3S1N11, LM3S1N16, LM3S1W16, LM3S1Z16, LM3S3826, LM3S3J26, LM3S3N26, LM3S3w26, LM3S5651, LM3S5656, LM3S5951, LM3S5956, LM3S5K31, LM3S5K36, LM3S5P31, LM3S5P36, LM3S5P51, LM3S5P56, LM3S5R31, LM3S5R36, LM3S5T36, LM3S5Y36, LM3S9997, and LM3S9L97.

8.1.2 Add support for Code Composer Studio (Reference 10407)

Support has been added for the Code Composer Studio tools from Texas Instruments. Compiler specific code has been added where necessary in order to work with the new compiler, and all example applications now have CCS project files in addition to the already existing project files for the other toolchains.

8.2 Bug Fixes in Stellaris Graphics Library

8.2.1 Added explicit casts in imgbutton.h (Reference 11149)

Explicit casts have been added to various macros defined in the ImageButton widget header file, imgbutton.h. These allow either a generic tWidget or specific tImageButtonWidget pointer to be passed as the first parameter without a compiler warning being generated.

8.3 New Features in Stellaris Utility Library

8.3.1 Baud rate used by uartstdio may now be specified (Reference 11075)

A new function, UARTStdioInitExpClk(), has been added to the uartstdio module API. This function takes two parameters, the UART port number and baud rate, and may be used instead of UARTStdioInit() for applications which wish to run the UART at baud rates other than the default 115,200bps.

8.4 New Features in Third Party Packages

8.4.1 Updated to lwIP 1.3.1 (Reference 10577)

lwIP 1.3.1 has been incorporated into StellarisWare and all example applications that use lwIP have been updated to use the new version. Additionally, the lwIP driver now has support for being used with SafeRTOS.

8.4.2 Added headers for accessing SafeRTOS (Reference 11070)

Header files have been added for use by applications that wish to utilize the SafeRTOS stored in the ROM of certain Stellaris devices.

8.4.3 Changed licensing terms on AES third party code (Reference 11176)

The AES implementation is now based on PolarSSL-0.10.1 which is covered by a BSD-style license. Previously the AES implementation was covered by the LGPL. See the file LICENSE.txt for the licensing terms.

8.5 New Features in DK-LM3S9B96 Firmware Package

8.5.1 Added SafeRTOS demo application (Reference 11008)

Added an example application (safertos_demo) that utilizes SafeRTOS and lwIP. The behavior of the SafeRTOS tasks can be modified via the web server provided via lwIP, and tasks can be created/terminated via the touch screen display.

8.5.2 enet_uip example added to dk-lm3s9b96 release (Reference 10961)

A new example application, enet_uip, has been added to the dk-lm3s9b96 release. This implements a very simple web server on top of the uIP TCP/IP stack.

8.6 Bug Fixes in DK-LM3S9B96 Firmware Package

8.6.1 Fixed flash programming error in usb_stick_update (Reference 11041)

The first two locations were being programmed as all Fs instead of being left unprogrammed during the initial programming of the image. This practice violates the programming rules from some parts. The example application has now been changed to leave the first two locations unprogrammed until the last step of image programming.

8.6.2 Removed unneeded data alignment pragmas for IAR tools (Reference 10993)

This only affects IAR tools. There were extra pragma statements like this: #pragma data_alignment=4, intended to reset the data alignment after earlier setting the alignment to 1024. This is actually not needed and these extra pragmas were causing warnings.

8.6.3 Support for lower audio rates (Reference 11107)

The sound driver provided with the dk-lm3s9b96 was not capable of supporting lower audio sample rates. This was because the MCLK source clock could not be divided down enough to support lower rates on the I2S interface. This change allows audio sample rates down to 6.1 kHz mono 8 bit audio. The lower sample rates are supported by shifting out more bits than are used for 8 and 16 bit audio formats. This solution may not be acceptable for I2S audio DACs that only support right justified I2S audio formats or require the number of bits per frame to exactly match the number of bits in the audio sample.

8.6.4 Added explicit casts in imgbutton.h (Reference 11149)

Explicit casts have been added to various macros defined in the ImageButton widget header file, imgbutton.h. These allow either a generic tWidget or specific tImageButtonWidget pointer to be passed as the first parameter without a compiler warning being generated.

8.6.5 Minor change in lwIP tcp.h header file (Reference 11218)

To work around an apparent bug in the IAR 5.41 compiler, four macros in the lwIP 1.3.1 header tcp.h have been replaced with simpler (but equivalent) versions. If the original definitions are used, IAR 5.41 generates code which is incorrect and results in TCP/IP connection ACK processing problems. This change will be removed once a compiler fix is available.

8.7 New Features in EK-LM3S3748 Firmware Package

8.7.1 LMScope now provides a helpful error if run without the driver installed (Reference 11099)

In previous releases, the LMScope Windows example application statically linked the USB driver DLL lmusbdll.dll. This meant that the user would see a generic Windows “DLL not found” message if the application was run before the USB device driver for the ek-lm3s3748 qs-scope device were installed. The application has now been reworked to dynamically load the driver DLL and this allows it to provide a more helpful message in cases where the driver doesn’t exist.

8.8 Bug Fixes in EK-LM3S3748 Firmware Package

8.8.1 Fixed flash programming error in usb_stick_update (Reference 11041)

The first two locations were being programmed as all Fs instead of being left unprogrammed during the initial programming of the image. This practice violates the programming rules from some parts. The example application has now been changed to leave the first two locations unprogrammed until the last step of image programming.

8.8.2 Removed unneeded data alignment pragmas for IAR tools (Reference 10993)

This only affects IAR tools. There were extra pragma statements like this: #pragma data_alignment=4, intended to reset the data alignment after earlier setting the alignment to 1024. This is actually not needed and these extra pragmas were causing warnings.

8.8.3 Added explicit casts in imgbutton.h (Reference 11149)

Explicit casts have been added to various macros defined in the ImageButton widget header file, imgbutton.h. These allow either a generic tWidget or specific tImageButtonWidget pointer to be passed as the first parameter without a compiler warning being generated.

8.9 Bug Fixes in EK-LM3S6965 Firmware Package

8.9.1 Minor change in lwIP tcp.h header file (Reference 11218)

To work around an apparent bug in the IAR 5.41 compiler, four macros in the lwIP 1.3.1 header tcp.h have been replaced with simpler (but equivalent) versions. If the original definitions are used, IAR 5.41 generates code which is incorrect and results in TCP/IP connection ACK processing problems. This change will be removed once a compiler fix is available.

8.10 Bug Fixes in EK-LM3S8962 Firmware Package

8.10.1 Minor change in lwIP tcp.h header file (Reference 11218)

To work around an apparent bug in the IAR 5.41 compiler, four macros in the lwIP 1.3.1 header tcp.h have been replaced with simpler (but equivalent) versions. If the original definitions are used, IAR 5.41 generates code which is incorrect and results in TCP/IP connection ACK processing problems. This change will be removed once a compiler fix is available.

8.11 New Features in EK-LM3S9B90 Firmware Package

8.11.1 Add new example using the uDMA controller with a periodic timer (Reference 11061)

A new example has been created to demonstrate use of DMA with a periodic timer. A uDMA channel is configured to perform a transfer based on a periodic timer timeout.

8.11.2 Add new example using the uDMA controller with a timer edge capture mode (Reference 11143)

A new example has been created to demonstrate use of DMA with a timer configured for an edge capture mode. A timer is configured for the edge timer capture mode. This mode is used to measure time between edges on an input pin. A uDMA channel is configured to transfer the captured time into a buffer and after a certain number of edges are captured the program analyzes the captured data. This example uses a PWM output as a stimulus and requires two signals to be jumpered together on the evaluation board.

8.12 Bug Fixes in EK-LM3S9B90 Firmware Package

8.12.1 Fixed flash programming error in usb_stick_update (Reference 11041)

The first two locations were being programmed as all Fs instead of being left unprogrammed during the initial programming of the image. This practice violates the programming rules from some parts. The example application has now been changed to leave the first two locations unprogrammed until the last step of image programming.

8.12.2 Removed unneeded data alignment pragmas for IAR tools (Reference 10993)

This only affects IAR tools. There were extra pragma statements like this: #pragma data_alignment=4, intended to reset the data alignment after earlier setting the alignment to 1024. This is actually not needed and these extra pragmas were causing warnings.

8.12.3 Minor change in lwIP tcp.h header file (Reference 11218)

To work around an apparent bug in the IAR 5.41 compiler, four macros in the lwIP 1.3.1 header tcp.h have been replaced with simpler (but equivalent) versions. If the original definitions are used, IAR 5.41 generates code which is incorrect and results in TCP/IP connection ACK processing problems. This change will be removed once a compiler fix is available.

8.13 New Features in EK-LM3S9B92 Firmware Package

8.13.1 Add new example using the uDMA controller with a periodic timer (Reference 11061)

A new example has been created to demonstrate use of DMA with a periodic timer. A uDMA channel is configured to perform a transfer based on a periodic timer timeout.

8.13.2 Add new example using the uDMA controller with a timer edge capture mode (Reference 11143)

A new example has been created to demonstrate use of DMA with a timer configured for an edge capture mode. A timer is configured for the edge timer capture mode. This mode is used to measure time between edges on an input pin. A uDMA channel is configured to transfer the captured time into a buffer and after a certain number of edges are captured the program analyzes the captured data. This example uses a PWM output as a stimulus and requires two signals to be jumpered together on the evaluation board.

8.14 Bug Fixes in EK-LM3S9B92 Firmware Package

8.14.1 Fixed flash programming error in usb_stick_update (Reference 11041)

The first two locations were being programmed as all Fs instead of being left unprogrammed during the initial programming of the image. This practice violates the programming rules from some parts. The example application has now been changed to leave the first two locations unprogrammed until the last step of image programming.

8.14.2 Removed unneeded data alignment pragmas for IAR tools (Reference 10993)

This only affects IAR tools. There were extra pragma statements like this: #pragma data_alignment=4, intended to reset the data alignment after earlier setting the alignment to 1024. This is actually not needed and these extra pragmas were causing warnings.

8.14.3 Minor change in lwIP tcp.h header file (Reference 11218)

To work around an apparent bug in the IAR 5.41 compiler, four macros in the lwIP 1.3.1 header tcp.h have been replaced with simpler (but equivalent) versions. If the original definitions are used, IAR 5.41 generates code which is incorrect and results in TCP/IP connection ACK processing problems. This change will be removed once a compiler fix is available.

8.15 Bug Fixes in RDK-ACIM Firmware Package

8.15.1 Fix temperature calculation for motor kits (Reference 11029)

Modify the calculation of ambient temperature to use a signed short value rather than an unsigned char value. This will correct the problem of temperature wrapping to a large number and triggering a temperature fault on the motor kit when the temperature drops below 0 degrees Celsius. This fix will update both the motor kit firmware and the motor kit GUI software.

8.16 New Features in RDK-BDC24 Firmware Package

8.16.1 Added RDK-BDC24 support (Reference 11170)

The firmware for the RDK-BDC24 motor controller has been added.

8.17 Bug Fixes in RDK-BLDC Firmware Package

8.17.1 Fix temperature calculation for motor kits (Reference 11029)

Modify the calculation of ambient temperature to use a signed short value rather than an unsigned char value. This will correct the problem of temperature wrapping to a large number and triggering a temperature fault on the motor kit when the temperature drops below 0 degrees Celsius. This fix will update both the motor kit firmware and the motor kit GUI software.

8.18 Bug Fixes in RDK-IDM Firmware Package

8.18.1 Added explicit casts in imgbutton.h (Reference 11149)

Explicit casts have been added to various macros defined in the ImageButton widget header file, imgbutton.h. These allow either a generic tWidget or specific tImageButtonWidget pointer to be passed as the first parameter without a compiler warning being generated.

8.18.2 Minor change in lwIP tcp.h header file (Reference 11218)

To work around an apparent bug in the IAR 5.41 compiler, four macros in the lwIP 1.3.1 header tcp.h have been replaced with simpler (but equivalent) versions. If the original definitions are used, IAR 5.41 generates code which is incorrect and results in TCP/IP connection ACK processing problems. This change will be removed once a compiler fix is available.

8.19 Bug Fixes in RDK-IDM-L35 Firmware Package

8.19.1 Added explicit casts in imgbutton.h (Reference 11149)

Explicit casts have been added to various macros defined in the ImageButton widget header file, imgbutton.h. These allow either a generic tWidget or specific tImageButtonWidget pointer to be passed as the first parameter without a compiler warning being generated.

8.20 Bug Fixes in RDK-IDM-SBC Firmware Package

8.20.1 Fixed flash programming error in usb_stick_update (Reference 11041)

The first two locations were being programmed as all Fs instead of being left unprogrammed during the initial programming of the image. This practice violates the programming rules from some parts.

The example application has now been changed to leave the first two locations unprogrammed until the last step of image programming.

8.20.2 Removed unneeded data alignment pragmas for IAR tools (Reference 10993)

This only affects IAR tools. There were extra pragma statements like this: #pragma data_alignment=4, intended to reset the data alignment after earlier setting the alignment to 1024. This is actually not needed and these extra pragmas were causing warnings.

8.20.3 Added explicit casts in imgbutton.h (Reference 11149)

Explicit casts have been added to various macros defined in the ImageButton widget header file, imgbutton.h. These allow either a generic tWidget or specific tImageButtonWidget pointer to be passed as the first parameter without a compiler warning being generated.

8.20.4 Minor change in lwIP tcp.h header file (Reference 11218)

To work around an apparent bug in the IAR 5.41 compiler, four macros in the lwIP 1.3.1 header tcp.h have been replaced with simpler (but equivalent) versions. If the original definitions are used, IAR 5.41 generates code which is incorrect and results in TCP/IP connection ACK processing problems. This change will be removed once a compiler fix is available.

8.21 Bug Fixes in RDK-S2E Firmware Package

8.21.1 Minor change in lwIP tcp.h header file (Reference 11218)

To work around an apparent bug in the IAR 5.41 compiler, four macros in the lwIP 1.3.1 header tcp.h have been replaced with simpler (but equivalent) versions. If the original definitions are used, IAR 5.41 generates code which is incorrect and results in TCP/IP connection ACK processing problems. This change will be removed once a compiler fix is available.

8.22 Bug Fixes in RDK-STEPPER Firmware Package

8.22.1 Fix temperature calculation for motor kits (Reference 11029)

Modify the calculation of ambient temperature to use a signed short value rather than an unsigned char value. This will correct the problem of temperature wrapping to a large number and triggering a temperature fault on the motor kit when the temperature drops below 0 degrees Celsius. This fix will update both the motor kit firmware and the motor kit GUI software.

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9.1 New Features in Stellaris Boot Loader

9.1.1 Added Internal Pull up/down to boot loader. (Reference 10876)

The boot loader has the ability to check an external pin to see if it is set to a high or low voltage level and remain in the boot loader. However, the boot loader did not have a way to enable an internal pull-up or pull-down when using this feature. Two additional configuration options were added to enable an internal pull-up resistor by defining FORCED_UPDATE_WPU or an internal pull-down by defining FORCED_UPDATE_WPD value. These values are mutually exclusive and only one or the other should be used when using an internal pull resistor. If an external resistor pull-up or pull-down is used then neither valued should be defined.

9.2 New Features in Stellaris Peripheral Driver Library

9.2.1 Added IntPendSet() and IntPendClear() APIs (Reference 10694)

The IntPendSet() and IntPendClear() APIs have been added to the interrupt controller driver. They allow an interrupt to be pended (made to appear as if it has occurred) or unpended (made to appear as if it has not occurred). Unpending an interrupt is sometimes useful prior to enabling it (if the process of configuring the interrupt source causes a false trigger prior to the interrupt being enabled).

9.3 Bug Fixes in Stellaris Peripheral Driver Library

9.3.1 I2CMasterErr returned I2C_MASTER_ERR_NONE if arbitration was lost (Reference 10821)

Function I2CMasterErr previously assumed that bit 2 of the I2CMCS register would be set in all error conditions and, if this bit was clear, assumed no error had occurred. Unfortunately, this bit only indicates an ACK error so the function would return I2C_MASTER_ERR_NONE if the controller lost arbitration. This has been fixed.

9.3.2 USBFIFOFlush() fails to flush endpoints. (Reference 10853)

The USBFIFOFlush() function was improperly checking the state of the FIFO and was not allowing endpoints to flush the endpoint's FIFO. This affected all endpoints other than endpoint zero.

9.3.3 Use of non-MOTO formats in SSIConfigSetExpClk() can cause minor errors in the actual clock rate (Reference 10922)

If a non-MOTO format was specified in a call to the SSIConfigSetExpClk() function, two lower bits of a clock divisor register could be corrupted. The result was a small error in the actual clock rate. This has been fixed.

9.3.4 Added support for PWM3 trigger to ADC (Reference 10943)

Add the ability to configure an ADC sample sequence to be triggered by the event generated by the PWM3 generator, when present. The ADC_TRIGGER_PWM3 is now supported by ADCSequenceConfigure().

9.3.5 Corrected definition of NUM_INTERRUPTS (Reference 10908)

The definition of NUM_INTERRUPTS was off by one, resulting in improper treatment of the last interrupt (GPIOJ) by the IntRegister()/IntUnregister() functions, and assertion failures by the remaining Interrupt driver APIs.

9.4 Bug Fixes in Stellaris Utility Library

9.4.1 Fixed trailing slash handling errors in fswrapper (Reference 10892)

Three related changes have been made to fix problems experienced when using fswrapper and the "cd" command on the qs-checkout application serial command line. Previously fswrapper would result in a NULL pointerin ppcFSFilename if the pcName string passed did not contain a second

slash character. It now returns a pointer to the terminating NULL instead, preventing a corrupt string being displayed by the application. Additionally, file.c in qs-checkout has been modified to ensure that it never leaves a trailing slash at the end of the current working directory string and will also detect and fail attempts to change into non-FAT directories since the internal file system image access functions do not support directory navigation and listing.

9.5 New Features in DK-LM3S9B96 Firmware Package

9.5.1 qs-checkout application updated to support FPGA/Camera daughter board (Reference 10647)

Minor changes have been made to the qs-checkout example application to ensure that it operates correctly with the FPGA/Camera daughter board attached. In this case, no external RAM is available to store a directly accessible copy of the photo gallery file system image so this web site is not available and the “Image Viewer” function reports “No file system image”.

9.6 Bug Fixes in DK-LM3S9B96 Firmware Package

9.6.1 JPEG decode example rebranding (Reference 10614)

The JPEG image used in the showjpeg example application has been replaced with a version including the Texas Instruments logo rather than the Luminary Micro logo.

9.6.2 Fixed trailing slash handling errors in fswrapper (Reference 10892)

Three related changes have been made to fix problems experienced when using fswrapper and the “cd” command on the qs-checkout application serial command line. Previously fswrapper would result in a NULL pointerin ppcFSFilename if the pcName string passed did not contain a second slash character. It now returns a pointer to the terminating NULL instead, preventing a corrupt string being displayed by the application. Additionally, file.c in qs-checkout has been modified to ensure that it never leaves a trailing slash at the end of the current working directory string and will also detect and fail attempts to change into non-FAT directories since the internal file system image access functions do not support directory navigation and listing.

9.6.3 Corrected error in IAR linker script for ext_demo_1 and ext_demo_2 (Reference 10978)

An error in the definition of the size of the SRAM on the Flash/SRAM/LCD daughter board was fixed in the IAR toolchain linker scripts for example applications ext_demo_1 and ext_demo_2. The previous versions defined the SRAM to be 64KB in size when it should have been 1MB.

9.6.4 eflash tool added to dk-lm3s9b96 release (Reference 10979)

Although documentation for the “eflash” tool was included in previous StellarisWare releases for dk-lm3s9b96, the actual source and binary for the tool was omitted. These files have now been included and can be found in the “tools” subdirectory after StellarisWare installation.

9.7 Bug Fixes in EK-LM3S6965 Rev A Firmware Package

9.7.1 Fix handling of invalid page in uip web server (Reference 10981)

Fixed a cut-paste error in the setting of the page length for an invalid page. Also modified the invalid page to return very simple “page not found” content.

9.8 Bug Fixes in EK-LM3S6965 Firmware Package

9.8.1 Fix handling of invalid page in uip web server (Reference 10981)

Fixed a cut-paste error in the setting of the page length for an invalid page. Also modified the invalid page to return very simple “page not found” content.

9.9 Bug Fixes in EK-LM3S8962 Firmware Package

9.9.1 Fix handling of invalid page in uip web server (Reference 10981)

Fixed a cut-paste error in the setting of the page length for an invalid page. Also modified the invalid page to return very simple “page not found” content.

9.10 Bug Fixes in EK-LM3S9B90 Firmware Package

9.10.1 Fix handling of invalid page in uip web server (Reference 10981)

Fixed a cut-paste error in the setting of the page length for an invalid page. Also modified the invalid page to return very simple “page not found” content.

9.11 Bug Fixes in EK-LM3S9B92 Firmware Package

9.11.1 Fix handling of invalid page in uip web server (Reference 10981)

Fixed a cut-paste error in the setting of the page length for an invalid page. Also modified the invalid page to return very simple “page not found” content.

9.12 Bug Fixes in RDK-BDC Firmware Package

9.12.1 Properly handle overflow in PID controller (Reference 8535)

An overflow of the output of the PID controller will now clip instead of wrapping. The wrapping behavior could cause unexpected behavior, like the motor being driven in the wrong direction. With clipping, the motor behaves as expected.

9.13 New Features in RDK-IDM Firmware Package

9.13.1 Display and touchscreen driver updated to support the ILI9328 controller (Reference 10894)

The display and touchscreen drivers for rdk-idm have been updated to support the new KWH028Q02-F02 display module. This features an ILI9328 display controller which is compatible with the ILI9325 found on the previous module, KWH028Q02-F05. Minor updates in these drivers ensure that all three display controller IDs are handled correctly allowing a single binary to run on IDMs equipped with any of the displays.

9.14 Bug Fixes in RDK-IDM-SBC Firmware Package

9.14.1 qs-blox web site now updates correctly using IE7 (Reference 10613)

Special-case JavaScript has been added to the web site served by the qs-blox example application to ensure that the dynamically updated fields are correctly displayed in Internet Explorer 7. This browser insists on interpreting the XML response from the board as plain text so the JavaScript which receives the request now constructs an XML document around the response text and parses out the individual fields from it.

9.14.2 JPEG decode example rebranding (Reference 10614)

The JPEG image used in the showjpeg example application has been replaced with a version including the Texas Instruments logo rather than the Luminary Micro logo.

9.14.3 QS-Blox web site now updates in IE (Reference 10738)

A workaround for an Internet Explorer problem has been added to the web site served by the qs-blox example application. This allows the page to update automatically without a manual reload being required. The previous version of the web site operated correctly on standards-compliant browsers.

9.14.4 Fixed trailing slash handling errors in fswrapper (Reference 10892)

Three related changes have been made to fix problems experienced when using fswrapper and the “cd” command on the qs-checkout application serial command line. Previously fswrapper would result in a NULL pointerin ppcFSFilename if the pcName string passed did not contain a second slash character. It now returns a pointer to the terminating NULL instead, preventing a corrupt string being displayed by the application. Additionally, file.c in qs-checkout has been modified to ensure that it never leaves a trailing slash at the end of the current working directory string and will also detect and fail attempts to change into non-FAT directories since the internal file system image access functions do not support directory navigation and listing.

9.15 Bug Fixes in RDK-S2E Firmware Package

9.15.1 Fix storage of baud rate in RFC2217 code (Reference 10839)

In RFC2217, the baud rate parameter is a four octet parameter, which arrives in network (big-endian) order. The code that saves this parameter data as it arrives has been corrected to properly store it in host (little-endian) order.

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10.1 Bug Fixes in Stellaris Boot Loader

10.1.1 Ensure vector table is not compressed in IAR boot loader builds (Reference 10345)

The file bl_link.icf was updated to ensure that the IAR tools never compress the relocated boot loader vector table and initialized data sections. Using version 5.3 of Embedded Workbench for ARM, compression of these sections was noted in some cases during development of a boot loader supporting execution from external flash and, since relocation is not handled using IAR's code, this caused the boot loader to crash when run.

10.1.2 Use read-modify-write when configuring pins and peripherals in the boot loader (Reference 10348)

The boot loader now uses read-modify-write when enabling or disabling peripherals and configuring pins. This ensures that any peripheral or pin that has been configured during an application-specific hook function will not be unintentionally disabled or reconfigured by the boot loader code.

10.2 New Features in Stellaris Peripheral Driver Library

10.2.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.2.2 Added SSIBusy() function (Reference 9606)

This function determines if the SSI peripheral is busy transmitting data.

10.2.3 EPIConfigNoModeSet renamed to EPIConfigGPModeSet (Reference 10247)

The DriverLib API function EPIConfigNoModeSet has been renamed EPIConfigGPModeSet to ensure that the terminology used in the driver matches the part datasheets. Similarly, several labels defined in epi.h have been renamed to ensure consistency with the microcontroller documentation. The previous function and labels defined in epi.h have been deprecated but macros have been added to the header file to ensure that existing code using the function will still build and operate.

10.2.4 EPINonBlockingWriteCount renamed EPIWriteFIFOCountGet (Reference 10248)

The previous API EPINonBlockingWriteCount() has been renamed EPIWriteFIFOCountGet() to clarify operation. All EPI writes are via the write FIFO. If space is available, they do not block, otherwise they block until they can be added. The previous API has been deprecated but is mapped to the new function via a macro in epi.h to ensure backwards compatibility.

10.2.5 EPI driver function additions (Reference 10064)

Several changes have been made to the EPI driver. New configuration parameter flags have been added to EPIConfigHB8Set() and EPIConfigSDRAMSet() to allow configuration of features found in the EPIHB8CFG2 and EPISDRAMCFG2 registers respectively.

10.2.6 CANMessageSet() now provides flag to enable FIFO mode. (Reference 10431)

The CANMessageSet() did not allow configuring a set of message objects as a FIFO for transmitting or receiving CAN messages. This change adds the MSG_OBJ_FIFO flag value to tag message objects as part of a FIFO and not the final entry in a FIFO. This allows multiple message objects to be linked together to transfer or receive more than 8 bytes at a time.

10.3 Bug Fixes in Stellaris Peripheral Driver Library

10.3.1 Remove uDMAIntStatus() and uDMAIntClear() APIs (Reference 10148)

The APIs that use the DMA_CHIS register have been deleted because this register is no longer available for use and has been removed from the data sheet.

10.3.2 I2C_MASTER_CMD_BURST_RECEIVE_ERROR_STOP definition corrected (Reference 10434)

The value of I2C_MASTER_CMD_BURST_RECEIVE_ERROR_STOP was incorrect; the correct value is now provided.

10.3.3 CanBitRateSet() was incorrectly checking requested bit rate inputs. (Reference 10439)

The function CanBitRateSet() function was incorrectly checking the limits for possible bit rates given an input clock rate. The correct checking is now in place as ASSERT() checks to allow the parameter checking code to be removed in non-DEBUG builds.

10.3.4 Error in epi.h address size definitions corrected (Reference 10461)

An error in epi.h has been corrected and affected example applications have been updated. Labels EPI_ADDR_PER_SIZE_512MB and EPI_ADDR_RAM_SIZE_512MB have been replaced with the correct definitions for those register bit patterns, EPI_ADDR_PER_SIZE_256MB and EPI_ADDR_RAM_SIZE_256MB.

10.4 New Features in Stellaris Host Tools

10.4.1 pnmtoc now supports grayscale “PGM” files (Reference 10402)

The pnmtoc tool used to convert PBM/PNM format images into C arrays for use with the Stellaris Graphics Library has been updated to support conversion of grayscale “Portable Gray Map” (identifier “P5”) images. The previous version supported only color PBM images with identifier “P6”. Gray map images are created by the open source giftopnm tool when the input GIF image contains only shades of gray.

10.4.2 Windows USB example application rebranding (Reference 10500)

The Windows USB examples and their installer have been changed to present TI branding rather than Luminary Micro. This rework involved no functional changes in any application. Applications now install into “C:\Program Files\Texas Instruments\Stellaris” by default and shortcuts can be found under “Start\All Programs\Texas Instruments\Stellaris”.

10.5 Bug Fixes in Stellaris USB Library

10.5.1 USB host applications using OTG mode will hang if connected as a device. (Reference 10142)

USB application that are using OTG mode without having initialized device mode operation will hang if they are connected as a device to another USB host. This bug will affect all USB host example applications on kits that provide a USB OTG connector.

10.6 Bug Fixes in Stellaris Utility Library

10.6.1 Additional pointer checking added to fswrapper (Reference 10374)

When using a position independent file system image, some additional safety checks have been added in fswrapper and the qs-checkout application for dk-lm3s9b96 to prevent a fault exception from occurring if the file system image is corrupted. The file system code now checks that pointers calculated from the linked list in the file system image are actually within the bounds of the image itself before dereferencing them. Although a corrupt file system is itself a serious error, this change allows the qs-checkout application to boot successfully even with a bad file system image in SSI flash and allows the user to update the bad image via TFTP.

10.7 Bug Fixes in Third Party Packages

10.7.1 JPEG decoder reworked to use new ExtRAMAlloc/Free functions. (Reference 10262)

The JPEG decoder has been reworked to use functions ExtRAMAlloc() and ExtRAMFree() in place of the now-deprecated SDRAMAlloc() and SDRAMFree().

10.8 New Features in DK-LM3S9B96 Firmware Package

10.8.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.8.2 Higher data rate audio files can now be played in qs-checkout (Reference 10151)

Recent optimizations in the USB library have allowed the previous 64KB/S data rate restriction when playing uncompressed audio files from a USB flash stick to be lifted. Smooth audio playback is now possible from USB flash sticks for stereo files at 44.1KHz and 48KHz while running the qs-checkout example application.

10.8.3 Support added for SRAM/Flash/LCD Daughter Board (Reference 10307)

Three new examples and a new driver have been added to the StellarisWare release for dk-lm3s9b96 in support of the new SRAM/Flash/LCD daughter board. The examples are a sample boot loader (boot_eth_ext) allowing images to be written to external flash and booted from there and two small example applications that will operate with that boot loader, ext_demo_1 and ext_demo_1. A new driver, extflash.c, is included to support erasing and programming the external flash.

10.8.4 EPIConfigNoModeSet renamed to EPIConfigGPModeSet (Reference 10247)

The DriverLib API function EPIConfigNoModeSet has been renamed EPIConfigGPModeSet to ensure that the terminology used in the driver matches the part datasheets. Similarly, several labels defined in epi.h have been renamed to ensure consistency with the microcontroller documentation. The previous function and labels defined in epi.h have been deprecated but macros have been added to the header file to ensure that existing code using the function will still build and operate.

10.8.5 PinoutSet() now performs dynamic EPI configuration. (Reference 10042)

Function PinoutSet() in drivers/set_pinout.c has been reworked to offer dynamic configuration of the Extended Peripheral Interface (EPI) based upon information read from an I2C-connected EEPROM device. The new code is intended to support future daughter cards for the dk-lm3s9b96 board which will feature this EEPROM device and is used in all dk-lm3s9b96 example applications by default. To allow simplified EPI and pinout configurations and to remove the overhead of the new dynamic configuration code, a second implementation of the PinoutSet() function is provided which merely configures the pinout and EPI based on a hardcoded configuration. This may be enabled by building the file with label SIMPLE_PINOUT_SET defined.

10.8.6 Embedded web site rework (Reference 10517)

The embedded web sites served by the enet_io and enet_lwip example applications have been reworked to use 8.3 filenames. As a result, these sites can now be copied to SDCard and served from there when using the enet_lwip application.

10.8.7 New example applications ext_demo_1 and ext_demo_2 added (Reference 9968)

Two new example applications illustrating execution from EPI-connected flash have been added to the dk-lm3s9b96 release. Both are intended for use with the SRAM/Flash/LCD daughter board and the external flash Ethernet boot loader (boot_eth_ext).

10.8.8 Added an example boot loader targeting external flash (Reference 9513)

Example application boot_eth_ext has been added to the dk-lm3s9b96 StellarisWare package. This is a version of the Ethernet boot loader which can be used to download and run applications linked to run from the address space of the external flash found on the optional SRAM/Flash/LCD daughter board.

10.8.9 Display driver updated to support SRAM/Flash/LCD daughter board (Reference 9511)

The kitronix320x240x16_ssd2119_8bit.c display driver for dk-lm3s9b96 has been updated to support the SRAM/Flash/LCD daughter board and dynamically switch between GPIO-based and EPI-based access to the display depending upon the hardware detected. The display type determination is made inside function PinoutSet() which can be found in file drivers/set_pinout.c.

10.9 Bug Fixes in DK-LM3S9B96 Firmware Package

10.9.1 Error in epi.h address size definitions corrected (Reference 10461)

An error in epi.h has been corrected and affected example applications have been updated. Labels EPI_ADDR_PER_SIZE_512MB and EPI_ADDR_RAM_SIZE_512MB have been replaced with the correct definitions for those register bit patterns, EPI_ADDR_PER_SIZE_256MB and EPI_ADDR_RAM_SIZE_256MB.

10.10 New Features in EK-LM3S811 Firmware Package

10.10.1 Added support for RIT display on new ek-lm3s811 boards (Reference 10106)

The display driver for the ek-lm3s811 board has been updated to support both the OSRAM display found on the existing boards and the RIT display use on newer revisions. Since the driver is no longer specific to the OSRAM display, it has been renamed from osram96x16x1 to display96x16x1 and all example applications have been updated accordingly. Macros are provided in the new display96x169x.h header file to ensure that the previous API calls will be correctly remapped to the new functions.

This change causes the code size of the display driver to increase slightly but the new driver offers the ability to build for either one or other display by defining either OSRAM_ONLY or RIT_ONLY via the project file or makefile when building. If neither label is defined, the driver will include support for both displays and will determine which is required at runtime.

10.11 Bug Fixes in EK-LM3S811 Firmware Package

10.11.1 Quickstart application rebranded (Reference 10397)

The opening graphic displayed by the qs_ek-lm3s811 application has been changed to show the Texas Instruments logo rather than the Luminary Micro logo.

10.12 New Features in EK-LM3S2965 Firmware Package

10.12.1 Add a CAN FIFO example. (Reference 10430)

Two new CAN examples were added that use FIFO mode to transfer data to and from the main board to the CAN device board. The can_fifo example is run on the main board and the can_device_fifo is run on the CAN device board. These applications demonstrates how to use the DriverLib CAN APIs to use multiple CAN message objects as a FIFO for transferring data.

10.13 New Features in EK-LM3S6965 Rev A Firmware Package

10.13.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.13.2 Embedded web site rework (Reference 10524)

The embedded web site served by the enet_lwip example application has been reworked to use 8.3 filenames. As a result, this site can now be copied to SDCard and served from there when using the enet_lwip application.

10.14 Bug Fixes in EK-LM3S6965 Rev A Firmware Package

10.14.1 Embedded web site rebranding (Reference 10538)

The web sites served by the enet_io and enet_ptpd example applications have been rebranded to show Texas Instruments logos and information rather than the Luminary Micro equivalents. In the process, filenames were also updated to conform to the 8.3 naming convention to ease use of the sites from SDCards rather than internal flash.

10.15 New Features in EK-LM3S6965 Firmware Package

10.15.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.15.2 Embedded web site rework (Reference 10523)

The embedded web sites served by the enet_ptpd, enet_io and enet_lwip example applications have been reworked to use 8.3 filenames. As a result, these sites can now be copied to SDCard and served from there when using the enet_lwip application.

10.16 New Features in EK-LM3S8962 Firmware Package

10.16.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.16.2 Add a CAN FIFO example. (Reference 10430)

Two new CAN examples were added that use FIFO mode to transfer data to and from the main board to the CAN device board. The can_fifo example is run on the main board and the can_device_fifo is run on the CAN device board. These applications demonstrates how to use the DriverLib CAN APIs to use multiple CAN message objects as a FIFO for transferring data.

10.16.3 Embedded web site rework (Reference 10523)

The embedded web sites served by the enet_ptpd, enet_io and enet_lwip example applications have been reworked to use 8.3 filenames. As a result, these sites can now be copied to SDCard and served from there when using the enet_lwip application.

10.17 New Features in EK-LM3S9B90 Firmware Package

10.17.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.17.2 Add uDMA support to the enet_uip example application (Reference 10196)

The enet_uip example has been modified to add support for using uDMA with the Ethernet controller. This change is meant to demonstrate how an application can use the uDMA controller with the Ethernet controller in a general application.

10.17.3 EPIConfigNoModeSet renamed to EPIConfigGPModeSet (Reference 10247)

The DriverLib API function EPIConfigNoModeSet has been renamed EPIConfigGPModeSet to ensure that the terminology used in the driver matches the part datasheets. Similarly, several labels

defined in epi.h have been renamed to ensure consistency with the microcontroller documentation. The previous function and labels defined in epi.h have been deprecated but macros have been added to the header file to ensure that existing code using the function will still build and operate.

10.17.4 Embedded web site rework (Reference 10525)

The embedded web sites served by the enet_ptpd and enet_lwip example applications have been reworked to use 8.3 filenames and present TI brand information.

10.18 New Features in EK-LM3S9B92 Firmware Package

10.18.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.18.2 Add uDMA support to the enet_uip example application (Reference 10196)

The enet_uip example has been modified to add support for using uDMA with the Ethernet controller. This change is meant to demonstrate how an application can use the uDMA controller with the Ethernet controller in a general application.

10.18.3 EPIConfigNoModeSet renamed to EPIConfigGPModeSet (Reference 10247)

The DriverLib API function EPIConfigNoModeSet has been renamed EPIConfigGPModeSet to ensure that the terminology used in the driver matches the part datasheets. Similarly, several labels defined in epi.h have been renamed to ensure consistency with the microcontroller documentation. The previous function and labels defined in epi.h have been deprecated but macros have been added to the header file to ensure that existing code using the function will still build and operate.

10.18.4 Embedded web site rework (Reference 10525)

The embedded web sites served by the enet_ptpd and enet_lwip example applications have been reworked to use 8.3 filenames and present TI brand information.

10.19 Bug Fixes in RDK-ACIM Firmware Package

10.19.1 Change motor kit GUI install and start menu locations (Reference 10547)

The motor kit GUI programs install location has been changed from “Luminary Micro” to “Texas Instruments/Stellaris” in the Program Files directory. The Windows start menu location has likewise been changed so that the programs now appear under “Texas Instruments/Stellaris”.

10.20 New Features in RDK-BLDC Firmware Package

10.20.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.21 Bug Fixes in RDK-BLDC Firmware Package

10.21.1 Change motor kit GUI install and start menu locations (Reference 10547)

The motor kit GUI programs install location has been changed from “Luminary Micro” to “Texas Instruments/Stellaris” in the Program Files directory. The Windows start menu location has likewise been changed so that the programs now appear under “Texas Instruments/Stellaris”.

10.22 New Features in RDK-IDM-SBC Firmware Package

10.22.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.22.2 Functions SDRAMAlloc and SDRAMFree have been renamed. (Reference 10268)

The SDRAM memory management functions provided in drivers/sdram.c have been renamed to ExtRAMAlloc and ExtRAMFree to match the similar functions provided by the dk-lm3s9b96 software

release. These functions are used by the shared JPEG decoder software so must match across all releases which use JPEG. The previous functions are deprecated but macro definitions in sdram.h will remap them to the new names, ensuring that existing code which uses them will continue to build and operate.

10.23 Bug Fixes in RDK-IDM-SBC Firmware Package

10.23.1 Documentation correction (Reference 10139)

The documentation for the idm-checkout example application in the rdk-idm-sbc release of StellarisWare incorrectly stated that the board comes with a web site image already stored in the serial flash device. This is not the case - the user must download this image using TFTP if desired. The documentation has been updated to reflect this.

10.24 New Features in RDK-S2E Firmware Package

10.24.1 Add new GPIOPinTypeEthernetLED API (Reference 10090)

Add new API, GPIOPinTypeEthernetLED, to GPIO driver in driverlib. In this function the default drive strength for the Ethernet LED pins is now 8ma. Modified all of the sample applications that were enabling the Ethernet LED pin functions “manually” to now use this new API.

10.24.2 Embedded web site rework (Reference 10528)

The embedded configuration web site for the ser2enet application has been reworked to present TI brand information and the factory default name for the module has been changed from “Luminary Micro Serial2Ethernet Module” to “TI Stellaris Serial2Ethernet Module”. No functional changes were made to the application.

10.25 Bug Fixes in RDK-STEPPER Firmware Package

10.25.1 Fix Stack Overflow in RDK-Stepper Application (Reference 10459)

Under some circumstances the stack could overflow in the qs-stepper application. The stack size has been increased to accommodate the additional required stack space.

10.25.2 Change motor kit GUI install and start menu locations (Reference 10547)

The motor kit GUI programs install location has been changed from “Luminary Micro” to “Texas Instruments/Stellaris” in the Program Files directory. The Windows start menu location has likewise been changed so that the programs now appear under “Texas Instruments/Stellaris”.

11 Release Notes for StellarisWare Revision 4905 (July 30, 2009)

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11.1 New Features in Stellaris Boot Loader

11.1.1 Improved boot loader performance for dk-lm3s9b96 (Reference 9842)

The boot loader was reworked to offer the ability to replace the low level flash sizing, erase and programming functions. For Tempest-class devices such as the lm3s9b96, the boot loader flash programming function has been replaced with a version which makes use of the flash write buffer, improving download performance dramatically compared to the previous release. Note that the ROM-based boot loaders for Tempest-class devices were already making use of the flash write buffer - this change only affects boot loaders built to run from flash or SRAM.

11.1.2 Boot loader now allows vector table to be initialized in SRAM (Reference 9993)

A new label, VTABLE_START_ADDRESS, was added to the list supported by bl_config.h to allow an application to specify that its vector table should be relocated to SRAM prior to the boot loader transfering control to it. This is intended to support applications running out of EPI-connected memory where the vector table at the beginning of the application image is not accessible by the NVIC.

11.2 Bug Fixes in Stellaris Peripheral Driver Library

11.2.1 Missing configuration options added for EPIConfigHB8Set and EPIConfigNoModeSet (Reference 9778)

Additional flags have been added for use in the ulConfig parameter passed to EPIConfigNoModeSet and EPIConfigHB8Set. These allow selection of word access mode and, for the HostBus8 case, allow the function of CS to be defined. Prior to this addition, direct access to an EPI configuration register was required to set these options.

11.2.2 USBDevEndpointConfigSet() does not properly configure isochronous endpoints (Reference 9856)

The USBDevEndpointConfigSet() had an incorrect conditional statement that causes isochronous endpoints to be configured incorrectly. This causes the endpoint to acknowledge incoming packets, which results in the host controller not transmitting any more packets to that endpoint.

11.3 Bug Fixes in Stellaris Graphics Library

11.3.1 Rendering of 1bpp and 4bpp compressed images (Reference 9642)

GrImageDraw() now properly displays 1bpp and 4bpp compressed images. Due to a pair of rounding errors, previously it would incorrectly display images that were not a multiple of 8 pixels wide (1bpp) or 2 pixels wide (4bpp).

11.4 Bug Fixes in Stellaris Host Tools

11.4.1 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not

specifically handled and defaulted to MIME type text/plain.

11.5 New Features in Stellaris USB Library

11.5.1 Add support for USB audio class in device mode. (Reference 9894)

This update adds support for USB audio device class to the USB library. The USB audio class support only audio playback with volume control implemented as well. The audio device class is implemented as a generic USB audio class and will work with any operating system that supports USB audio class devices without the need for additional operating system drivers.

11.6 Bug Fixes in Stellaris USB Library

11.6.1 USB library enumeration code not properly clearing FIFO flags. (Reference 10044)

When handling standard device request, the USB library was not always clearing the FIFO read flag when a new request was received. This could effect any commands that followed the request that failed to clear the FIFO read flag.

11.6.2 USB library not handling VBUS errors in OTG mode. (Reference 10100)

The USB library is not properly handling VBUS error interrupts during host enumeration in OTG mode. If VBUS error occurs during enumeration, it causes the USB library code to hang in an intermediate state and not properly turn off power and retry the connection.

11.7 New Features in Stellaris Utility Library

11.7.1 New function ustrnicmp added to the ustdlib module (Reference 9862)

An implementation of the standard C runtime strnicmp (compare strings without regard to case) has been added to the ustrlib module.

11.7.2 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of

transfering files to and from an application over Ethernet.

11.8 Bug Fixes in Stellaris Utility Library

11.8.1 Fix UDP-Only configuration of LWIP (Reference 9898)

Add conditional compilation wrapper to the code that supports the TCP timer. This will prevent undefined references when compiling the LWIP library for UDP only (i.e. no TCP).

11.9 Bug Fixes in Third Party Packages

11.9.1 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

11.10 New Features in DK-LM3S9B96 Firmware Package

11.10.1 Improved boot loader performance for dk-lm3s9b96 (Reference 9842)

The boot loader was reworked to offer the ability to replace the low level flash sizing, erase and programming functions. For Tempest-class devices such as the lm3s9b96, the boot loader flash programming function has been replaced with a version which makes use of the flash write buffer, improving download performance dramatically compared to the previous release. Note that the ROM-based boot loaders for Tempest-class devices were already making use of the flash write buffer - this change only affects boot loaders built to run from flash or SRAM.

11.10.2 qs-checkout example now uses general-purpose TFTP server (Reference 9976)

The qs-checkout example application for dk-lm3s9b96 has been updated to make use of the new, general-purpose TFTP module (utils/tftp.c).

11.10.3 EPI configuration moved to PinoutSet() function (Reference 10012)

Configuration of the External Peripheral Interface (EPI) has been moved from individual drivers into the PinoutSet() function. This ensures that any external memories attached via EPI are accessible from early in the application startup process.

11.10.4 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of transfering files to and from an application over Ethernet.

11.11 Bug Fixes in DK-LM3S9B96 Firmware Package

11.11.1 Missing configuration options added for EPIConfigHB8Set and EPIConfigNoModeSet (Reference 9778)

Additional flags have been added for use in the ulConfig parameter passed to EPIConfigNoModeSet and EPIConfigHB8Set. These allow selection of word access mode and, for the HostBus8 case, allow the function of CS to be defined. Prior to this addition, direct access to an EPI configuration register was required to set these options.

11.11.2 Sound driver improperly calls buffer callback function. (Reference 10010)

The sound driver could call a buffer's callback function after the buffer had already been used. The buffer callback will now only be called if the buffer pointer is still valid.

11.11.3 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

11.11.4 Fix compatibility problem with recent Keil compiler in the usb_stick_update example application (Reference 10038)

The static declaration on a function was removed to work around a problem in the recent Keil toolchain (3.50) that was causing an internal fault in the compiler.

11.12 Bug Fixes in EK-LM3S3748 Firmware Package

11.12.1 Fix compatibility problem with recent Keil compiler in the usb_stick_update example application (Reference 10038)

The static declaration on a function was removed to work around a problem in the recent Keil toolchain (3.50) that was causing an internal fault in the compiler.

11.13 New Features in EK-LM3S6965 Firmware Package

11.13.1 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of transfering files to and from an application over Ethernet.

11.14 Bug Fixes in EK-LM3S6965 Firmware Package

11.14.1 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

11.15 New Features in EK-LM3S8962 Firmware Package

11.15.1 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of transfering files to and from an application over Ethernet.

11.16 Bug Fixes in EK-LM3S8962 Firmware Package

11.16.1 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

11.17 New Features in EK-LM3S9B90 Firmware Package

11.17.1 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of transfering files to and from an application over Ethernet.

11.18 Bug Fixes in EK-LM3S9B90 Firmware Package

11.18.1 Missing configuration options added for EPIConfigHB8Set and EPIConfigNoModeSet (Reference 9778)

Additional flags have been added for use in the ulConfig parameter passed to EPIConfigNoModeSet and EPIConfigHB8Set. These allow selection of word access mode and, for the HostBus8 case, allow the function of CS to be defined. Prior to this addition, direct access to an EPI configuration register was required to set these options.

11.18.2 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

11.18.3 Fix compatibility problem with recent Keil compiler in the usb_stick_update example application (Reference 10038)

The static declaration on a function was removed to work around a problem in the recent Keil toolchain (3.50) that was causing an internal fault in the compiler.

11.19 New Features in EK-LM3S9B92 Firmware Package

11.19.1 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of transfering files to and from an application over Ethernet.

11.20 Bug Fixes in EK-LM3S9B92 Firmware Package

11.20.1 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

11.20.2 Fix compatibility problem with recent Keil compiler in the usb_stick_update example application (Reference 10038)

The static declaration on a function was removed to work around a problem in the recent Keil toolchain (3.50) that was causing an internal fault in the compiler.

11.21 New Features in RDK-IDM Firmware Package

11.21.1 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of transfering files to and from an application over Ethernet.

11.22 Bug Fixes in RDK-IDM Firmware Package

11.22.1 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

11.23 New Features in RDK-IDM-SBC Firmware Package

11.23.1 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of transfering files to and from an application over Ethernet.

11.23.2 Add USB Memory Stick Updater Application (Reference 10048)

Two new applications have been added to demonstrate the ability to perform a firmware update over USB from a memory stick. The application usb_stick_update is the updater, and the application usb_stick_demo provides an example that can be loaded from a USB memory stick.

11.24 Bug Fixes in RDK-IDM-SBC Firmware Package

11.24.1 Sound driver improperly calls buffer callback function. (Reference 10010)

The sound driver could call a buffer's callback function after the buffer had already been used. The buffer callback will now only be called if the buffer pointer is still valid.

11.24.2 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

11.25 New Features in RDK-S2E Firmware Package

11.25.1 General purpose TFTP server module added (Reference 10053)

A new general purpose TFTP server module has been added to the utils directory for all kits supporting Ethernet. This module makes use of the lwIP TCP/IP stack and offers a simple method of transfering files to and from an application over Ethernet.

11.26 Bug Fixes in RDK-S2E Firmware Package

11.26.1 MIME type for icons is now correctly set (Reference 10021)

The lwIP HTTPD server and makefsfile tool have both been updated to correctly recognize files with extension ".ico" and report these as type image/x-icon. In previous releases, these were not specifically handled and defaulted to MIME type text/plain.

12 Release Notes for StellarisWare Revision 4781 (June 30, 2009)

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12.1 New Features in Stellaris Peripheral Driver Library

12.1.1 Add API for ADC Digital Comparators (Reference 9668)

In the Tempest class devices, a Digital Comparator feature has been added to the ADC module. This feature allows ADC samples to be sent to a digial comparator. This comparator can be programmed to trigger on low-band, mid-band or high-band values, and the trigger can be used to generate an interrupt or trigger a fault condition to the PWM module. The ADC API has been expanded to provide support for this new feature. Additional details about the operation of the digital comparator can be found in the data sheets for the Tempest Class Stellaris devices.

12.1.2 Added support devices that support 32 USB endpoints. (Reference 9666)

Newer devices allow for more endpoints with up to 32 IN/OUT endpoints now available. Support for the additional endpoints was added to the DriverLib USB functions and examples of using the new APIs were added to the USB library. The main change to the DriverLib API was to deprecate the interrupt handling functions because they could not support 32 endpoints as defined. The deprecated APIs are USBIntDisable(), USBIntEnable(), USBIntStatus() which can still be used with older devices that have only 8 endpoints. There are six added APIs that provide the same functionality, except the new APIs are split between control interrupts and endpoint interrupts. The new APIs are the following: USBIntDisableControl(), USBIntEnableControl(), USBIntStatusControl(), USBIntDisableEndpoint(), USBIntEnableEndpoint(), and USBIntStatusEndpoint(). The flags to use with the new USB control interrupt functions start with USB_INTCTRL_ while the new USB endpoint interrupt functions use the USB_INTEP_ flags.

12.2 Bug Fixes in Stellaris Peripheral Driver Library

12.2.1 SysCtlClockGet() provided incorrect results in some cases (Reference 9555)

If the PLL is enabled, SysCtlClockGet() now applies the system divider to the computed PLL output frequency even if the USESYSDIV bit in RCC is not set. It is possible to configure RCC where the PLL is enabled and USESYSDIV is not set, but the device forces the use of the system divider (since the PLL is enabled). This change causes SysCtlClockGet() to mimic the behavior of the device and therefore provide correct results in this case.

12.2.2 USBDevEndpointConfig() deprecated in favor of USBDevEndpointConfigSet() (Reference 9297)

The function USBDevEndpointConfig() has been marked as DEPRECATED and the name has been changed to USBDevEndpointConfigSet() to be symmetrical with the USBDevEndpointConfigGet() API. This has no affect on any current code however the definitions for USBDevEndpointConfig() may be removed at some point in the future.

12.2.3 SysCtlPeripheralPresent() did not properly handle USB (Reference 9756)

The SysCtlPeripheralPresent() API added a case to handle the USB peripheral because the current definition will incorrectly indicate the presence of the USB0 peripheral even when there is no USB controller present.

12.3 Bug Fixes in Stellaris Graphics Library

12.3.1 WidgetRemove() now clears the widget's next pointer (Reference 9615)

The next pointer on a widget is now cleared when it is removed from the widget tree with WidgetRemove(). If the widget is later added back to the widget tree with WidgetAdd(), it will no longer corrupt the widget tree since the next pointer no longer points to a potentially invalid widget.

12.3.2 GrStringSet() did not properly handle the ulSize parameter (Reference 9630)

The GrStringGet() was not using the ulSize parameter in all cases and was allowing the function to write beyond the end of the buffer provided to the function. This could cause other variables or data to overwritten with data for a given string.

12.4 New Features in DK-LM3S9B96 Firmware Package

12.4.1 Add USB Memory Stick Updater Application (Reference 9722)

Two new applications have been added to demonstrate the ability to perform a firmware update over USB from a memory stick. The application usb_stick_update is the updater, and the application usb_stick_demo provides an example that can be loaded from a USB memory stick.

12.5 New Features in EK-LM3S3748 Firmware Package

12.5.1 Add USB Memory Stick Updater Application (Reference 9722)

Two new applications have been added to demonstrate the ability to perform a firmware update over USB from a memory stick. The application usb_stick_update is the updater, and the application usb_stick_demo provides an example that can be loaded from a USB memory stick.

12.6 New Features in EK-LM3S9B90 Firmware Package

12.6.1 Add USB Memory Stick Updater Application (Reference 9722)

Two new applications have been added to demonstrate the ability to perform a firmware update over USB from a memory stick. The application usb_stick_update is the updater, and the application usb_stick_demo provides an example that can be loaded from a USB memory stick.

12.7 New Features in EK-LM3S9B92 Firmware Package

12.7.1 Add USB Memory Stick Updater Application (Reference 9722)

Two new applications have been added to demonstrate the ability to perform a firmware update over USB from a memory stick. The application usb_stick_update is the updater, and the application usb_stick_demo provides an example that can be loaded from a USB memory stick.

12.8 Bug Fixes in RDK-IDM Firmware Package

12.8.1 sd_card application was not properly configuring the Graphics Library (Reference 9793)

The sd_card application was failing to properly configure the Graphics Library before calling GrStringDraw() which was causing the application to halt. This problem occurred whenever a re-

quest to update the firmware was received from the Ethernet controller.

12.9 Bug Fixes in RDK-IDM-SBC Firmware Package

12.9.1 Corrected text misalignment in usb_host_mouse and usb_host_keyboard (Reference 9787)

The text strings in the status panel at the bottom of the display for the IDM-SBC versions of usb_host_mouse and usb_host_keyboard have been moved to prevent possible overlap.

12.10 Bug Fixes in Stellaris Firmware Development Package

12.10.1 Updated project files for Sourcery G++ for Stellaris (Reference 9667)

The project files for Sourcery G++ for Stellaris have been updated to reflect the requirements of the new version of CodeSourcery's tool chain. This fixes some of the project/workspace import problems that occurred with the previous version of StellarisWare and Sourcery G++ for Stellaris.

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13.1 New Features in Stellaris Graphics Library

13.1.1 Additions to the ImageButton widget (Reference 9484)

New functionality has been added to the ImageButton widget offered by the Graphics Library. The widget now supports IB_STYLE_FILL for drawing a background color and new macros allow the button background and keycap images to be enabled and disabled. These changes have been implemented to keep the interface backwards compatible, hence the use of style flags IB_STYLE_KEYCAP_OFF and IB_STYLE_IMAGE_OFF (since the previous version of the widget assumed that both images were disabled unless a NULL pointer was used to populate the relevant image pointer).

13.2 New Features in Stellaris USB Library

13.2.1 USB host event driver added to USB library (Reference 9534)

An event driver was added to the USB host library to provide notification of important system events and class specific events that were previously not visible to the application.

13.3 Bug Fixes in DK-LM3S9B96 Firmware Package

13.3.1 i2s_demo application report wrong elapsed time for 8bit wav files. (Reference 8973)

There was a problem in the interpretation of the byte rate of .wav files being played that caused the byte rate calculation to be incorrect by a factor of 2 for 8 bit .wav files.

14 Release Notes for StellarisWare Revision 4674 (May 19, 2009)

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14.1 Bug Fixes in Stellaris Boot Loader

14.1.1 Ethernet boot loader hangs in some cases (Reference 9240)

A delay has been added between enabling the Ethernet controller and accessing it. If the boot loader was configured to enable the Ethernet LEDs, this was performing the function of that delay. If the LEDs were not used, a fault would occur since the Ethernet controller was accessed too quickly after being enabled. This delay resolves that problem in the case that the LEDs are not used.

14.2 New Features in Stellaris Peripheral Driver Library

14.2.1 Added two new uDMA API functions to support the interrupt status register in Tempest (Reference 9179)

Added two new functions to the uDMA API: uDMAIntStatus() and uDMAIntClear() to support the new DMA interrupt status register that is available in Tempest class parts.

14.2.2 Add CAN Bit Rate API (Reference 9315)

A simplified CAN Bit Rate API, CANBitRateSet() was added to provide an easier method of setting the CAN bit timing as opposed to having to fully specify the CAN bit timing with the CANBitTimingSet() API. The new API can directly set the CAN bit rate based on the clock source for the CAN controller. The CANBitTimingSet() API can still be used if more precise timing parameters are needed by an application.

14.2.3 Added I2S and EPI drivers to DriverLib (Reference 9419)

Drivers have been added for the I2S and EPI peripherals that are available on the new Stellaris parts.

14.3 Bug Fixes in Stellaris Graphics Library

14.3.1 Corrected operation of ListBoxLock() and ListBoxUnlock() (Reference 9441)

In previous releases, the operation of the ListBoxLock() and ListBoxUnlock() macros was reversed. This has now been corrected.

14.3.2 Corrected operation of Lock and Unlock macros for Slider and JPGWidget (Reference 9471)

In previous releases, the operation of the SliderLock()/SliderUnlock() and JPEGWidgetLock()/JPEGWidgetUnlock() macros were reversed. This has now been corrected.

14.4 New Features in Stellaris Host Tools

14.4.1 New board locater tool for Ethernet-based applications (Reference 9094)

A board locater tool has been created that will search the Ethernet network for Stellaris-based boards running code which utilizes Ethernet and the board locater service. This allows an easy method to discover the presence, IP address, and MAC address of the Ethernet-based boards on the network, as well as a description of the application that is running on that board.

14.4.2 Windows USB Examples have moved to the tools directory (Reference 9388)

The Windows USB example applications which were previously found in the StellarisWare/boards/ek-lm3s3748/windows_examples directory have moved to the StellarisWare/tools directory instead. Previously, these examples only applied to the lm3s3748 kit but, with the introduction of new lm3s9b90, lm3s9b92 and lm3s9b96 kits, they are now required by several boards so this change ensures that they are in a single, common location for all boards that make use of them.

14.4.3 Update to Red Suite Project Import XML Files (Reference 9445)

Red Suite Version 2 upgrades the compiler tools to version 4.3.2. For building the boot loader, the -Os option should be specified now (same as for Code Sourcery and GCC). Also, the way that compiler defines are specified has changed and the new XML files will support these changes.

14.5 Bug Fixes in Stellaris Host Tools

14.5.1 Makefsfile updated to prevent generation of invalid C code (Reference 8651)

The makefsfile tool was updated to ensure that filenames containing characters which are not legal within C variable names would be correctly translated into something that is valid C. In the previous version, only spaces, dots and slashes were replaced with underscores. The new version adds all the non-alphanumeric (shifted) characters to this list. Without this change, it was possible to generate a C file system image file which would not compile if filenames in the directory being imaged contained characters such as ‘-’, ‘+’.

14.5.2 makefsfile tool now adds correct headers to XML files (Reference 9361)

The makefsfile tool, used to generate images for internal file systems which can be used by the lwIP HTTP server, would previously describe any file with a '.xml' extension as 'text/plain' rather than 'text/xml'. This caused problems when using AJAX since the XMLHttpRequest object would not have the responseXML field set when the asynchronous request completed (the response would be stored only in the responseText field).

14.5.3 Library files for Windows USB DLLs have been added to the release (Reference 9386)

In previous releases, the library files lmusbdll.lib and lmdfu.lib were missing making it impossible to build some of the USB example applications without having access to the Windows Device Driver Kit. These files have now been added to the appropriate directories under StellarisWare/tools allowing the examples to be built. Additionally, copies of the files have been included in the Windows device driver package (SW-USB-windrivers) which already includes the DLLs that these library files relate to, lmusbdll.dll and lmdfu.dll.

14.6 Bug Fixes in Stellaris USB Library

14.6.1 Fixed a USB host MSC bug causing a hang on multi-block reads (Reference 9411)

A bug existed in the previous release which would cause USBHMSCBlockRead() and USBHMSCBlockWrite() to hang if passed a ulNumBlocks value greater than 1. This was due to an assumption in usbhscsi.c that all reads and writes would be performed on a block-by-block basis resulting in incorrect block numbers being written to the read and write command blocks generated in USBHSCSIRead10() and USBHSCSIWrite10().

14.6.2 USBDCDInit() now disconnects before reconnecting (Reference 9442)

The USBLib device initialization function USBDCDInit() now explicitly disconnects the device from the bus and delays approximately 100mS before connecting it once again. The previous version of the function did not perform this disconnect operation and, as a result, if the function was called when the device was already connected to the USB bus, it would not be reenumerated resulting in missing callbacks to the application and resulting application confusion.

14.7 New Features in Stellaris Utility Library

14.7.1 Added function fs_map_path() to fswrapper module (Reference 9322)

The fswrapper module offers a method to give multiple file system images user-friendly names in web URLs. It can support FAT logical drives and binary file system images but only provides the subset of file system operations typically required by a web server. To allow access to the more advanced functions provided by FatFS for FAT logical drives, a new API has been provided, fs_map_path(), which will map a path in the fswrapper namespace to the equivalent path at the FatFS level (for mount points that correspond to FAT logical drives). For example, passing the string "/sdcard/index.htm" would return "0:/index.htm" assuming the mount point name "sdcard" is associated with FAT logical drive number 0.

14.8 Bug Fixes in Stellaris Utility Library

14.8.1 Change between static and DHCP IP sometimes fails (Reference 9438)

The function, lwIPNetworkConfigChange, does not always switch properly between static IP and Auto IP (with DHCP). This is due to the fact that the variable that retains the current IP mode setting is not properly saved. This variable, g_ulIPMode, is now saved at the end of the function for all cases.

14.9 New Features in Third Party Packages

14.9.1 Added support for AES ROM tables in Tempest class parts (Reference 9089)

Modified the AES code in third_party to use the AES tables from ROM for Tempest class devices. Also modified the AES example applications for Tempest based boards.

14.10 Bug Fixes in Third Party Packages

14.10.1 Closed lwIP HTTPD timing hole that could cause hangs on connection shutdown (Reference 9256)

A race condition in the lwIP HTTPD server which could cause a NULL pointer to be dereferenced in some cases during connection termination was fixed.

14.10.2 lwIP HTTP server now sends correct headers for XML files (Reference 9358)

The lwIP HTTPD server previously described XML files using header “text/plain”. This caused problems for AJAX browser applications since the XML responses were not parsed correctly when received. The server now uses the correct “text/xml” header with any file whose extension is “.xml”.

14.11 New Features in DK-LM3S9B96 Firmware Package

14.11.1 Added support for AES ROM tables in Tempest class parts (Reference 9089)

Modified the AES code in third_party to use the AES tables from ROM for Tempest class devices. Also modified the AES example applications for Tempest based boards.

14.12 Bug Fixes in DK-LM3S9B96 Firmware Package

14.12.1 Library files for Windows USB DLLs have been added to the release (Reference 9386)

In previous releases, the library files lmusbdll.lib and lmdfu.lib were missing making it impossible to build some of the USB example applications without having access to the Windows Device Driver Kit. These files have now been added to the appropriate directories under StellarisWare/tools allowing the examples to be built. Additionally, copies of the files have been included in the Windows device driver package (SW-USB-windrivers) which already includes the DLLs that these library files relate to, lmusbdll.dll and lmdfu.dll.

14.12.2 Corrected operation of Lock and Unlock macros for Slider and JPGWidget (Reference 9471)

In previous releases, the operation of the SliderLock()/SliderUnlock() and JPEGWidgetLock()/JPEGWidgetUnlock() macros were reversed. This has now been corrected.

14.12.3 Web server opens Luminary Micro site in the wrong frame (Reference 9488)

In various applications supporting an embedded web server (depending upon the kit, enet_io, enet_lwip, qs-checkout and idm-checkout) used to open the link to `http://www.luminarymicro.com` within a frame. The sites have been updated to open this link in the top level window instead.

14.13 New Features in EK-LM3S3748 Firmware Package

14.13.1 Windows USB Examples have moved to the tools directory (Reference 9388)

The Windows USB example applications which were previously found in the StellarisWare/boards/ek-lm3s3748/windows_examples directory have moved to the StellarisWare/tools directory instead. Previously, these examples only applied to the lm3s3748 kit but, with the introduction of new lm3s9b90, lm3s9b92 and lm3s9b96 kits, they are now required by several boards so this change ensures that they are in a single, common location for all boards that make use of them.

14.14 Bug Fixes in EK-LM3S3748 Firmware Package

14.14.1 Library files for Windows USB DLLs have been added to the release (Reference 9386)

In previous releases, the library files lmusbdll.lib and lmdfu.lib were missing making it impossible to build some of the USB example applications without having access to the Windows Device Driver Kit. These files have now been added to the appropriate directories under StellarisWare/tools allowing the examples to be built. Additionally, copies of the files have been included in the Windows device driver package (SW-USB-windrivers) which already includes the DLLs that these library files relate to, lmusbdll.dll and lmdfu.dll.

14.14.2 Stack overflow in usb_dev_serial example (Reference 9446)

The stack size allocated for the usb_dev_serial example was increased to prevent an overflow which had been seen occasionally in a previous version of the application.

14.14.3 Bitband example was failing to run on all tool chains. (Reference 9443)

The bitband example was failing on some tool chains due to the stack not being large enough. The stack size was increased to prevent the stack overflow from causing the application to crash.

14.15 Bug Fixes in EK-LM3S6965 Rev A Firmware Package

14.15.1 enet_ptpd web server occasionally returns too much data (Reference 9435)

The file system module in the enet_ptpd application used strlen() to determine the amount of data that should be served up by the web server instead of the file size that is stored in the file system structure. Return the stored size instead since the strlen() size since the later can be incorrect at times (if there is no trailing NULL in the file data).

14.15.2 Replace use of strstr with ustrstr (Reference 9447)

To avoid potential runtime library issues that vary from toolchain to toolchain, replace the use of the strstr function with ustrstr, which is provided in the utils folder.

14.16 Bug Fixes in EK-LM3S6965 Firmware Package

14.16.1 enet_ptpd web server occasionally returns too much data (Reference 9435)

The file system module in the enet_ptpd application used strlen() to determine the amount of data that should be served up by the web server instead of the file size that is stored in the file system structure. Return the stored size instead since the strlen() size since the later can be incorrect at times (if there is no trailing NULL in the file data).

14.16.2 Replace use of strstr with ustrstr (Reference 9447)

To avoid potential runtime library issues that vary from toolchain to toolchain, replace the use of the strstr function with ustrstr, which is provided in the utils folder.

14.17 Bug Fixes in EK-LM3S8962 Firmware Package

14.17.1 enet_ptpd web server occasionally returns too much data (Reference 9435)

The file system module in the enet_ptpd application used strlen() to determine the amount of data that should be served up by the web server instead of the file size that is stored in the file system structure. Return the stored size instead since the strlen() size since the later can be incorrect at times (if there is no trailing NULL in the file data).

14.17.2 Replace use of strstr with ustrstr (Reference 9447)

To avoid potential runtime library issues that vary from toolchain to toolchain, replace the use of the strstr function with ustrstr, which is provided in the utils folder.

14.18 New Features in EK-LM3S9B90 Firmware Package

14.18.1 Added applications for new evaluation board (Reference 9348)

A suite of example applications has been added for the new evaluation board.

14.18.2 Added support for AES ROM tables in Tempest class parts (Reference 9089)

Modified the AES code in third_party to use the AES tables from ROM for Tempest class devices. Also modified the AES example applications for Tempest based boards.

14.19 Bug Fixes in EK-LM3S9B90 Firmware Package

14.19.1 Library files for Windows USB DLLs have been added to the release (Reference 9386)

In previous releases, the library files lmusbdll.lib and lmdfu.lib were missing making it impossible to build some of the USB example applications without having access to the Windows Device Driver Kit. These files have now been added to the appropriate directories under StellarisWare/tools allowing the examples to be built. Additionally, copies of the files have been included in the Windows device driver package (SW-USB-windrivers) which already includes the DLLs that these library files relate to, lmusbdll.dll and lmdfu.dll.

14.20 New Features in EK-LM3S9B92 Firmware Package

14.20.1 Added applications for new evaluation board (Reference 9348)

A suite of example applications has been added for the new evaluation board.

14.20.2 Added support for AES ROM tables in Tempest class parts (Reference 9089)

Modified the AES code in third_party to use the AES tables from ROM for Tempest class devices. Also modified the AES example applications for Tempest based boards.

14.21 Bug Fixes in EK-LM3S9B92 Firmware Package

14.21.1 Library files for Windows USB DLLs have been added to the release (Reference 9386)

In previous releases, the library files lmusbdll.lib and lmdfu.lib were missing making it impossible to build some of the USB example applications without having access to the Windows Device Driver Kit. These files have now been added to the appropriate directories under StellarisWare/tools allowing the examples to be built. Additionally, copies of the files have been included in the Windows device driver package (SW-USB-windrivers) which already includes the DLLs that these library files relate to, lmusbdll.dll and lmdfu.dll.

14.22 Bug Fixes in RDK-BLDC Firmware Package

14.22.1 Enhance Hall Sensor Speed Calculation (Reference 9476)

Modify the speed calculation algorithm to use every rising/falling edge of a Hall sensor input, instead of just the rising edge of Hall Sensor A. This improves the granularity of the speed calculation, and allows the PI loop to respond to changes in speed more quickly.

14.23 Bug Fixes in RDK-IDM-SBC Firmware Package

14.23.1 Corrected operation of Lock and Unlock macros for Slider and JPG-Widget (Reference 9471)

In previous releases, the operation of the SliderLock()/SliderUnlock() and JPEGWidget-Lock()/JPEGWidgetUnlock() macros were reversed. This has now been corrected.

14.23.2 Web server opens Luminary Micro site in the wrong frame (Reference 9488)

In various applications supporting an embedded web server (depending upon the kit, enet_io, enet_lwip, qs-checkout and idm-checkout) used to open the link to

`http://www.luminarymicro.com` within a frame. The sites have been updated to open this link in the top level window instead.

14.24 Bug Fixes in RDK-S2E Firmware Package

14.24.1 Change between static and DHCP IP sometimes fails (Reference 9438)

The function, lwIPNetworkConfigChange, does not always switch properly between static IP and Auto IP (with DHCP). This is due to the fact that the variable that retains the current IP mode setting is not properly saved. This variable, g_ulIPMode, is now saved at the end of the function for all cases.

14.25 New Features in Stellaris Firmware Development Package

14.25.1 Add SourceryG++ for Stellaris project files (Reference 9469)

Project files (.sgxx) and workspace files (.sgxw) for use by the SourceryG++ for Stellaris IDE are now provided for the libraries, applications, and boards that are provided in StellarisWare.

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